NEET 2021

OFFICIAL QUESTION PAPERS WITH DETAILED SOLUTION

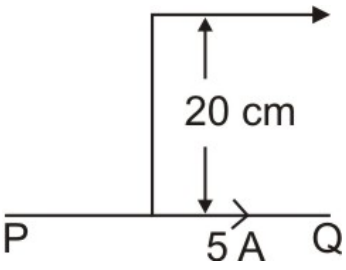
All questions with detailed solutions of Physics, Chemistry and Biology

CAREERS 360

Physics

Q. 1 An infinitely long straight conductor carries a current of 5A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant.
Calculate the magnitude of the force experienced by the electron at that instant.

$\text{Electron} v = 10^5\ m/s$



Option 1:
$4 \times 10^{-20} N$

Option 2:
$8\pi \times 10^{-20} N$

Option 3:
$4\pi \times 10^{-20} N$

Option 4:
$8 \times 10^{-20} N$

Correct Answer:
$8 \times 10^{-20} N$

Solution:
$i = 5A$

$e = 1.6 \times 10^{-19} c\ \ v = 10^5 m/s$

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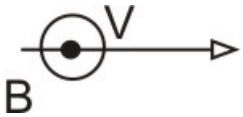
$$F = qvB\ \sin\theta = qvB\sin 90^o$$

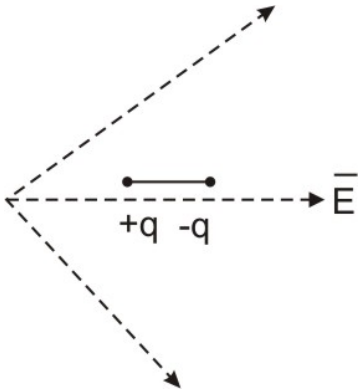


$$B = \frac{\mu_o I}{2\pi r}$$

$$F = 1.6 \times 10^{-19} \times 10^5 \times \frac{2 \times 10^{-7} \times 5}{2 \times 10^{-1}}$$

$$F = 8 \times 10^{-20} N$$

Q. 2 A dipole is placed in an electric field as shown. In which direction will it move?



Option 1:
towards the left as its potential energy will increase.

Option 2:
towards the right as its potential energy will decrease.

Option 3:
towards the left as its potential energy will decrease.

Option 4:
towards the right as its potential energy will increase.

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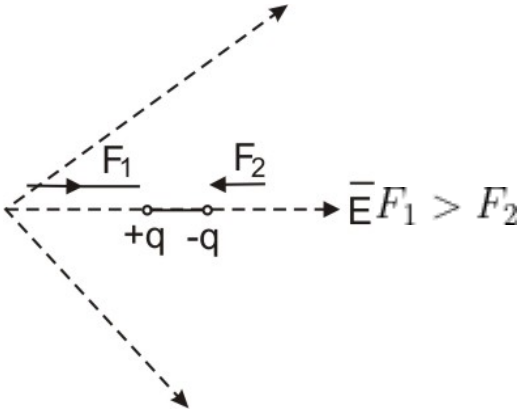
Correct Answer:
towards the right as its potential energy will decrease.

Solution:



$F_1 > F_2$

F_{net}(towards right)

electric field intensity is more at $+q$ as field lines are closer

$V_2 - V_1 = -PE_2\cos\theta_2 - (-PE_1\cos\theta_1)$

$\theta_1 = \theta_2 = 180^o$(along givenline)

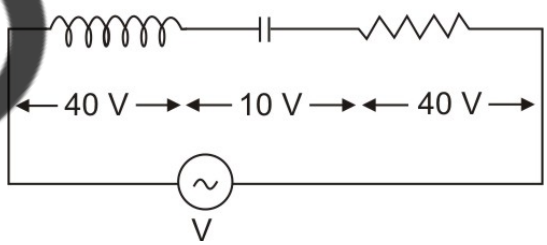
$E_2 < E_1$(as it move towards right)

$= (+PE_2) - (+PE_1)$

$\therefore \Delta U \Rightarrow Decrease$

Q. 3 An inductor of inductance L, a capacitor of capacitance C and a resistor of resistance 'R' are connected in series to an ac source of potential difference 'V' volts as shown in figure.

Potential difference across, C and R is 40 V, 10V and 40V, respectively. The amplitude of current flowing through LCR series circuit is $10\sqrt{2}A$. The impedance of the circuit is:



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Option 1:
$4\sqrt{2}\Omega$

Option 2:
$\frac{5}{\sqrt{2}}\,\Omega$

Option 3:
$4\,\Omega$

Option 4:
$5\,\Omega$

Correct Answer:
$5\,\Omega$

Solution:

$$V_{rms} = \sqrt{V_R^2 + (V_L - V_c)^2}$$

$$= \sqrt{(40)^2 + (30)^2} = 50$$

$$V_{peak} = (V_{rms}) \times \sqrt{2}$$

$$= 50\sqrt{2}$$

$$Z = \frac{V_{peak}}{I_{peak}} = \frac{50\sqrt{2}}{10\sqrt{2}} = 5\Omega$$

Q. 4 For a plane electromagnetic wave propagating it x-direction, which one of the following combination given the correct possible direction for electric field (E) and magnetic field (B) respectively?

Option 1:
$\hat{j} + \hat{k}, \hat{j} + \hat{k}$

Option 2:
$-\hat{j} + \hat{k}, -\hat{j} - \hat{k}$

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Option 3:

$\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

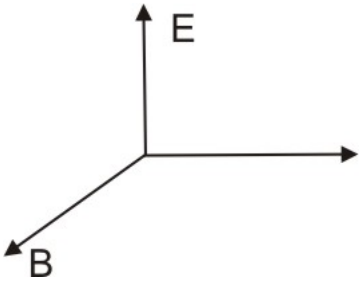
Option 4:

$-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$

Correct Answer:

$-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

Solution:



Direction of proparation $\bar{E}, \bar{B}\&$ dirction of proparation are mutually perpendicular to each other

$$\bar{E} \cdot \bar{B} = EB\cos 90^o = 0$$

Q. 5 A lens of large focal length and large aperture is best suited as an objective of an astronomic telescope since:

Option 1:

A large aperture contributes to the quality and visibility of image.

Option 2:

A large area of the objective ensures better light gathering power.

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Option 3:
a large aperture provides a better resolution

Option 4:
all of the above

Correct Answer:
all of the above

Solution:

Magnification of telescope$= \frac{f_o}{\mu_o} = f_o\left[\frac{1}{fe} + \frac{1}{Ve}\right]$

Large focal length$\longrightarrow$better Magnification

Large aperture$\longrightarrow$better light gathering power

Hence, the answer is the option (4).

Q. 6 **Column - I** gives certain physical terms associated with the flow of current through a metallic conductor.
Column - II gives some mathematical relations involving electrical quantities.
Match Column - I and Column - II with appropriate relation.

	Column I		Column II
(A)	Drift Velocity	(P)	$\frac{m}{ne^2p}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaction Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

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Option 1:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Option 2:
(A)-(R), (B)-(S), (C)-(Q), (D)-(P)

Option 3:
(A)-(R), (B)-(P), (C)-(S), (D)-(Q)

Option 4:
(A)-(R), (B)-(Q), (C)-(S), (D)-(P)

Correct Answer:
(A)-(R), (B)-(S), (C)-(P), (D)-(Q)

Solution:

$$\frac{E}{J}=\frac{1}{\sigma}=\rho=Electrical Resistivity$$

Driff Velocity $=V_2=\frac{eE\tau}{m}$

Relaxation Period $=\tau=\frac{m}{ne^2\varrho}$

Current density$=J=neV_2$

$$A-R, B-S, C-P, D-Q$$

Hence, the answer is the option (1).

Q. 7 An electromagnetic wave of wavelength $'\lambda'$ is incident on a photosensitive surface of negligible work function. If 'm' mass is of photoelectron emitted from the surface has de-Broglie wavelength λ_d, then:

Option 1:

$$\lambda=\left(\frac{2m}{hc}\right)\lambda_d{}^2$$

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Option 2:

$$\lambda_d = \left(\frac{2mc}{h}\right)\lambda^2$$

Option 3:

$$\lambda = \left(\frac{2mc}{h}\right)\lambda_d{}^2$$

Option 4:

$$\lambda = \left(\frac{2h}{mc}\right)\lambda_d{}^2$$

Correct Answer:

$$\lambda = \left(\frac{2mc}{h}\right)\lambda_d{}^2$$

Solution:

Work Function$\rightarrow$negligible

$$\phi \simeq 0$$

$$\frac{hc}{\lambda} = \phi + kE_{max}$$

$$\frac{hc}{\lambda} = 0 + \frac{p^2}{2m} = \frac{(h/\lambda d)^2}{2m}$$

$$\frac{kc}{\lambda} = \frac{h}{\lambda_d^2 \times 2m}$$

$$\left(\frac{2mc}{h}\right)\lambda_d^2 = \lambda$$

Hence, the answer is the option (3).

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Q. 8 A particle is released from height S from the surface of the earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the earth and the speed of the particle at that instant are respectively:

Option 1:
$\frac{S}{4}, \frac{3gS}{2}$

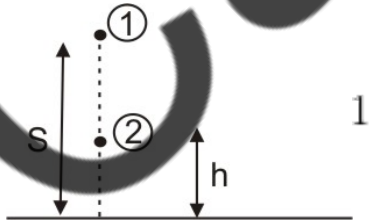
Option 2:
$\frac{S}{4}, \frac{\sqrt{3gS}}{2}$

Option 3:
$\frac{S}{2}, \frac{\sqrt{3gS}}{2}$

Option 4:
$\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

Correct Answer:
$\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

Solution:



$1 \rightarrow Initial$

$2 \rightarrow Final$

$TE_1 = TE_2$

$mgs + 0 = (mgh) + KE$

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$(kE) = 3\,(mgh)\,(Given)$

$mgs = 4mgh$

$h = \frac{s}{4}$

$3mgh = \frac{1}{2}mv^2$

$v = \sqrt{2gh \times 3} = \sqrt{2g\frac{s}{4} \times 3} = \sqrt{\frac{gs}{2} \times 3}$

$v = \sqrt{\frac{3gS}{2}}$

Q. 9 A screw gauge gives the following reading when used to measure the dimeter of a wire

Main scale reading : 0 mm

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is:

Option 1:
0.52 cm

Option 2:
0.026 cm

Option 3:
0.26 cm

Option 4:
0.052 cm

Correct Answer:
0.052 cm

Solution:

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$1\ MSD = 100CSD = 1mm$

$CSR = 52Division = 0.52mm$

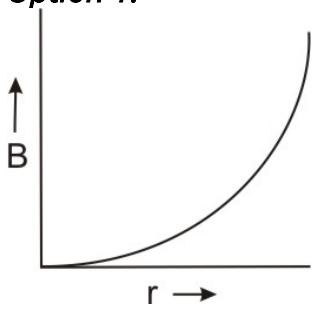
$Reading = MSR + CSR$

$= 0 + 0.52mm$

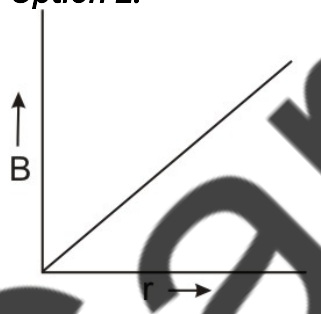
$= 0.052cm$

Q. 10 A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field B(r) due to the cable with the distance 'r' from the axis of the cable is represented by:

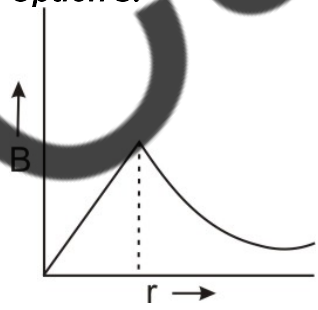
Option 1:



Option 2:



Option 3:



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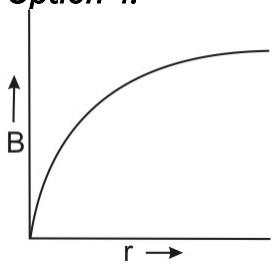
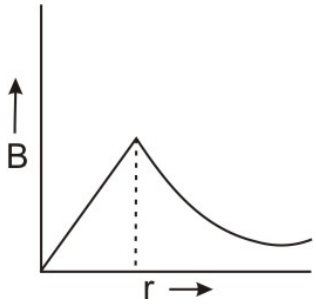
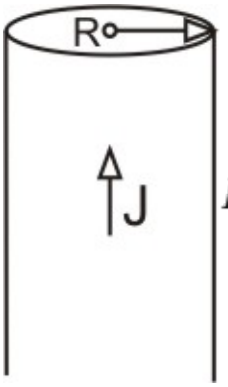
Option 4:



Correct Answer:

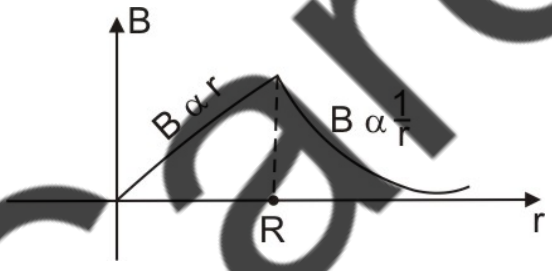


Solution:



$$B = \frac{\mu_o I r}{2\pi R^2} \, (r < R)$$

$$B = \frac{\mu_o I}{2\pi r} \, (r > R)$$



Q. 11 Two charged spherical conductors of radius $R_1 \; and \; R_2$ are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

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Option 1:

$\frac{R_1}{R_2}$

Option 2:

$\frac{R_2}{R_1}$

Option 3:

$\sqrt{\left(\frac{R_1}{R_2}\right)}$

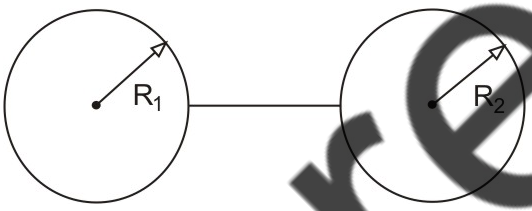
Option 4:

$\frac{{R_1}^2}{{R_2}^2}$

Correct Answer:

$\frac{R_2}{R_1}$

Solution:



$V_1 = V_2$

$\frac{kQ_1}{R_1} = \frac{kQ_2}{R_2}$

$\frac{Q_1}{Q_2} = \frac{R_1}{R_2}$

$\frac{\sigma_1}{\sigma_2} = \frac{Q_1/4\pi R_1^2}{Q_2/4\pi R_2^2} = \frac{R_2^2}{R_1^2} \times \frac{R_1}{R_2}$

$\frac{\sigma_1}{\sigma_2} = \frac{R_2}{R_1}$

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Q. 12 The number of photons per second on average emitted by the source of monochromatic light of wavelength 600 nm, when it delivers the power of $3.3 \times 10^{-3} watt$ will be :$\left(h = 6.6 \times 10^{-34} Js\right)$

Option 1:
10^{18}

Option 2:
10^{17}

Option 3:
10^{16}

Option 4:
10^{15}

Correct Answer:
10^{16}

Solution:
$\lambda = 600nm$

$P = 3.3 \times 10^{-3} w$

$h = 6.6 \times 10^{-34} Js$

$$\text{No. of photons per second} = \frac{p}{\frac{hc}{\lambda}} = \frac{3.3 \times 10^{-3}}{\frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{6 \times 10^{-7}}} = 1 \times 10^{16}$$

Hence, the answer is the option (3).

Q. 13 A cup of coffee cools from $90°C \; to \; 80°C$ in t minutes, when the room temperature is $20°C$. The time taken by a similar cup of coffee to cool from $80°C \; to \; 60°C$ at a room temperature same at $20°C$ is:

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Option 1:

$\frac{13}{10}t$

Option 2:

$\frac{13}{5}t$

Option 3:

$\frac{10}{13}t$

Option 4:

$\frac{5}{13}t$

Correct Answer:

$\frac{13}{5}t$

Solution:

$$\frac{\Delta T}{\Delta t} = -k\left(T_{ang} - T_0\right)$$

$$\frac{\frac{90-80}{t}}{\frac{(80-60)}{\Delta t}} = \frac{\left(\frac{90+80}{2} - 20\right)}{\left(\frac{80+60}{2} - 20\right)}$$

$$\frac{\frac{10}{t}}{\frac{20}{\Delta t}} = \frac{65}{50} = \frac{13}{10}$$

$$\frac{\Delta t}{2t} = \frac{13}{10}$$

$$\Delta t = \frac{13\,t}{5}$$

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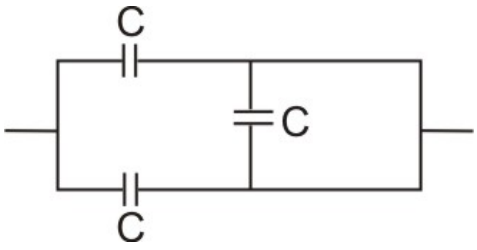
Q. 14 The equivalent capacitance of the combination shown in the figure is:



Option 1:
$3C$

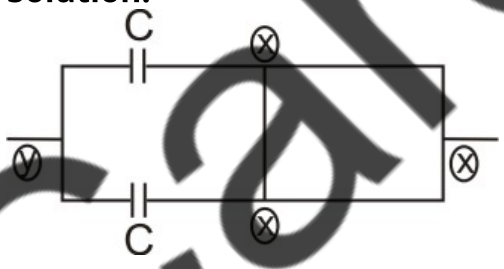
Option 2:
$2C$

Option 3:
$C/2$

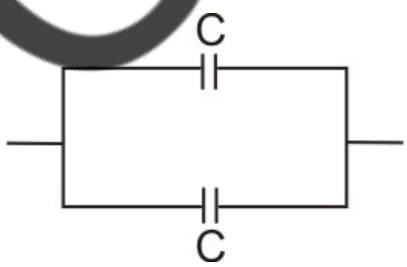
Option 4:
$3C/2$

Correct Answer:
$C/2$

Solution:



Capacitor is short circuited



$C_{eq} = 2C$

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Q. 15 The effective resistance of a parallel connection consists of four wires of equal length, equal area of cross-section, and the same material 0.25Ω. What will be the effective resistance if they are connected in series?

Option 1:
0.25Ω

Option 2:
0.5Ω

Option 3:
1Ω

Option 4:
4Ω

Correct Answer:
4Ω

Solution:
Four wires each of resistance for R parallel connection

$$R_{eq} = \frac{R}{4} = 0.25$$

$$\therefore R = 1$$

For series connection

$$R_{eq} = 4R = 4(1) = 4\Omega$$

Hence, the answer is the option (4).

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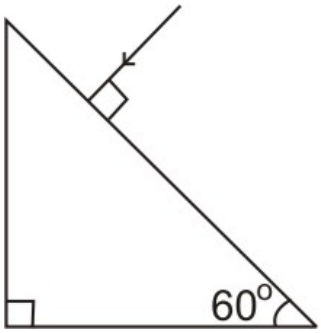
Q. 16 Find the value of the angle of emergency from the prism. Refractive index of the glass is $\sqrt{3}$.



Option 1:
60°

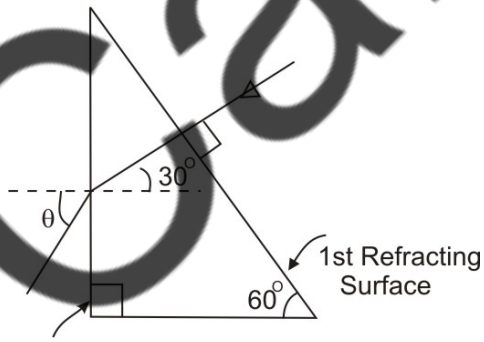
Option 2:
30°

Option 3:
45°

Option 4:
90°

Correct Answer:
60°

Solution:



2nd Refracting Surface

$$\mu_1 \sin 30^o = \mu_2 \sin \theta$$

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$$\sqrt{3} \times \frac{1}{2} = 1 \sin\theta$$

$$\theta = 60^o$$

$\theta = Angle\ of\ ehergence = 60^o$

Q. 17 A small block slides down on a smooth inclined plane, starting from rest at time t=0. Let Sn be the distance travelled by the block in the interval t=n-1 to t=n.

Then, the ratio $\frac{S_n}{S_{n+1}}$ is:

Option 1:

$\frac{2n-1}{2n}$

Option 2:

$\frac{2_n-1}{2n+1}$

Option 3:

$\frac{2_n+1}{2n-1}$

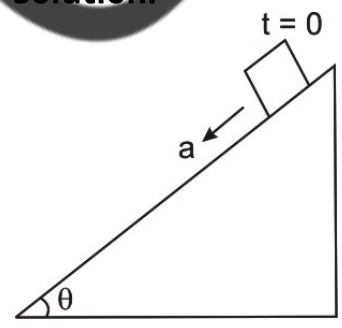
Option 4:

$\frac{2_n}{2n-1}$

Correct Answer:

$\frac{2_n-1}{2n+1}$

Solution:



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From $t = 0$ to $t = n - 1$

$$S_1 = 0 + \frac{1}{2}a\,(n-1)^2$$

From $t = 0\ to\ t = n$

$$S_2 = 0 + \frac{1}{2}a\,(n)^2$$

From $t = 0\ to\ t = n + 1$

$$S_3 = 0 + \frac{1}{2}a\,(n+1)^2$$

$$S_n = S_2 - S_1$$

$$S_n = \frac{1}{2}a\left[n^2 - \left(n^2 - 2n + 1\right)\right]$$

$$S_{n+1} = S_3 - S_2 = \frac{1}{2}a\left[n^2 + 2n + 1 - n^2\right]$$

$$\frac{S_n}{S_n + 1} = \frac{(2n-1)}{(2n+1)}$$

Q. 18 The escape velocity from the Earth's surface is V. The escape velocity from the surface of another planet having a radius, four times that of Earth and the same mass density is :

Option 1:
V

Option 2:
2V

Option 3:
3V

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Option 4:
4V

Correct Answer:
4V

Solution:

Escape velocity from the Earth's surface $= V = \sqrt{\frac{2GM}{R}}$

Escape velocity from a planet $= V' = \sqrt{\frac{2GM'}{R'}}$

$R' = 4R$

$M = 3 \times \frac{4}{3}\pi R^3$

$M' = 3 \times \frac{4}{3}\pi (R')^3 = 64M$

$\frac{V}{V'} = \sqrt{\frac{M}{M'} \times \frac{R'}{R}} = \sqrt{\frac{1}{64} \times \frac{4R}{R}}$

$\frac{V}{V'} = \frac{1}{4}$

$V' = 4V$

Hence, the answer is the option (4).

Q. 19 The half - life of a radioactive nuclide is 100 hours. The fraction of original activity that will remain after 150 hours would be:

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{2\sqrt{2}}$

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Option 3:

$\frac{2}{3}$

Option 4:

$\frac{2}{3\sqrt{2}}$

Correct Answer:

$\frac{1}{2\sqrt{2}}$

Solution:

$T_{\frac{1}{2}} = 100 hours$

$\frac{N_t}{N_o} = 2^{-t/\frac{T}{\frac{1}{2}}} = 2^{-150/100}$

$\frac{N_t}{N_o} = 2^{-3/2} = \frac{1}{2\sqrt{2}}$

Q. 20 In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replace the first cell, then at what length of the wire, the balance point occurs?

Option 1:

60 cm

Option 2:

21.6 cm

Option 3:

64 cm

Option 4:

62 cm

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Correct Answer:
60 cm

Solution:

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{1.5}{2.5} = \frac{36}{l_2}$$

$$\frac{3}{5} = \frac{36}{l_2} \qquad l_2 = 60Cm$$

Q. 21 Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? ($g = 10m/s^2$)

Option 1:
10.2 kW

Option 2:
8.1 kW

Option 3:
12.3 kW

Option 4:
7.0 kW

Correct Answer:
8.1 kW

Solution:

$$\frac{dm}{dt} = 15\frac{kg}{s}$$

$$\frac{P_{out}}{P_{input}} = \eta = \frac{90}{100}$$

$$P_{output} = \eta P_{input}$$

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$$= 0.9 \times \left(\frac{dmgh}{dt}\right)$$

$$= 0.9 \times 15 \times 10 \times 60$$

$$P_{output} = 8.1kW$$

Hence, the answer is the option (2).

Q. 22 The electron concentration in an n-type semiconductor is the same as the hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them.

Option 1:
Current in n-type = current in p-type

Option 2:
Current in p-type > current in n-type

Option 3:
Current in n-type < current in p-type

Option 4:
No current will flow in p-type ,current will only flow in n-type

Correct Answer:
Current in n-type = current in p-type

Solution:
In both n-type and p-type semiconductors, electric current is due to a minority charge carrier when an electric field is applied across each of them. (as there is no diffusion)

Current in n-type =current in p-type

Hence, the answer is the option (1).

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Q. 23 Mathch Column-I and Column-II and Choose the correct match from the given choice

Column-I	Column-II
(A) Root Mean square speed of gas molecules	(P) $\frac{1}{3}nm$
(B) Pressure exerted by ideal gas	(Q) $\sqrt{\frac{3RT}{M}}$
(C) Average kinetic energy of a molecules	(R) $\frac{5}{2}RT$
(D) Total internal energy of 1 mole of a diatomic gas	(S) $\frac{3}{2}KgT$

Option 1:
(A)-(R),(B)-(P),(C)-(S),(D)-(Q)

Option 2:
(A)-(Q),(B)-(R),(C)-(S),(D)-(P)

Option 3:
(A)-(Q),(B)-(P),(C)-(S),(D)-(R)

Option 4:
(A)-(R),(B)-(Q),(C)-(P),(D)-(S)

Correct Answer:
(A)-(Q),(B)-(P),(C)-(S),(D)-(R)

Solution:

Root Mean square speed of gas molecules=$V_{rms}=\sqrt{\frac{3RT}{M}}$

The pressure exerted by the ideal gas $=P=\frac{1}{3}nm\left(\bar{V}\right)^2$

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Average KE of a molecules$= \frac{3}{2}K_BT$

Total internal energy of 1 mole of a diatomic gas $= \frac{5}{2}RT$

Hence, the answer is the option (3).

Q. 24 A spring is stretched by 5 cm by a force of 10 N. The period of the oscillations when a mass of 2 kg is suspended by it is :

Option 1:
0.0628 s

Option 2:
6.28 s

Option 3:
3.14 s

Option 4:
0.628 s

Correct Answer:
0.628 s

Solution:

$x = 5cm$

$F = 10N$

$K = F/x$

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{200}}$

$T = \frac{\pi}{s} = 0.628s$

Hence, the answer is the option (4).

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Q. 25 The velocity of a small ball of mass M and density d,when dropped in a container filled with glycerine becomes constant after some time.if the density of glycerine is $\frac{d}{2}$,then the viscous force acting on the ball will be:

Option 1:
$\frac{Mg}{2}$

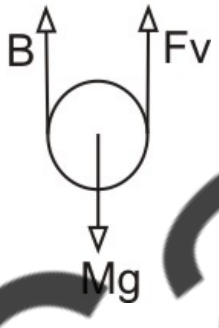
Option 2:
Mg

Option 3:
$\frac{3}{2}Mg$

Option 4:
$2Mg$

Correct Answer:
$\frac{Mg}{2}$

Solution:



$Mg = dvg$

$B = \frac{d}{2} \times vg = \frac{Mg}{2}$

$F_v \rightarrow viscous\ force$

$B + F_v = Mg$

$F_v = Mg - \frac{mg}{2}$

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$$F_v = \frac{mg}{2}$$

Q. 26 A body is executing simple harmonic motion with frequency 'n'. The frequency of its potential energy is:

Option 1:
n

Option 2:
$2n$

Option 3:
$3n$

Option 4:
$4n$

Correct Answer:
$2n$

Solution:

$x = A\sin\omega t$

$$PE = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2A^2\left[\frac{1-\cos 2\omega t}{2}\right]$$

$$pE = \frac{1}{4}m\omega^2A^2[1-\cos(20\omega t)]$$

As, $\omega = 2\pi n$

$$PE = \frac{1}{4}m\omega^2A^2[1-\cos(4\pi nt)]$$

The angular frequency of PE is 2ω

$\therefore$ Frequency=$2n$

Hence, the answer is the option (2).

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Q. 27 A nucleus with mass number 240 breaks into two fragments each of mass number 120, the binding energy per nucleon of unfragmented nuclei is 7.6 MeV while that of fragments is 8.5 MeV. The total gain the binding energy in the process is:

Option 1:
$0.9\ MeV$

Option 2:
$9.4MeV$

Option 3:
$804\ MeV$

Option 4:
$216\ MeV$

Correct Answer:
$216\ MeV$

Solution:
BE per nucleon of unfragmented nuclei $= 7.6MeV$

BE per nucleon of fragmented nuclei$= 8.5MeV$

An increase in BE per nucleon is $0.9MeV$

For 246 nucleons, total BE is $0.9 \times 240 = 216MeV$

Hence, the answer is the option (4).

Q. 28 A parallel plate capacitor has a uniform electric field $\vec{E}$ in the space between the plates. If the distance between the plates is 'd' and the area of each plate 'A', then the energy stored in the capacitor is :(ϵ_0 =permittivity of free space)

Option 1:
$\frac{1}{2}\epsilon_0 E^2$

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Option 2:
$\epsilon_0 EAd$

Option 3:
$\frac{1}{2}\epsilon_0 E^2 Ad$

Option 4:
$\frac{E^2 Ad}{\epsilon_0}$

Correct Answer:
$\frac{1}{2}\epsilon_0 E^2 Ad$

Solution:

$V = Ed, \quad C = \frac{A\varepsilon_0}{d}$

Energy stored in the capacitor $= \frac{1}{2}CV^2 = \frac{1}{2} \times \frac{A\varepsilon_0}{d} \times E^2 d^2 = \frac{1}{2}\varepsilon_0 E^2 Ad$

Hence, the answer is the option (3).

Q. 29 Polar molecules are the molecules:

Option 1:
having zero dipole moment .

Option 2:
acquire a dipole moment only in the presence of electric field due to displacement of charges.

Option 3:
acquire a dipole moment only when magnetic field is absent.

Option 4:
Having a permanent electric dipole moment .

Correct Answer:
Having a permanent electric dipole moment .

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Solution:
Polar molecules have permanent electric dipole moments.

Hence, the answer is the option (4).

Q. 30 A radioactive nucleus ${}^{A}_{Z}X$ undergoes spontaneous decay in the sequence ${}^{A}_{Z}X \rightarrow Z-1B \rightarrow Z-3 \rightarrow C \rightarrow Z-2D$,where Z is the atomic number of element X.The possible decay particles in the sequence are

Option 1:
α, β^-, β^+

Option 2:
α, β^+, β^-

Option 3:
β^+, α, β^-

Option 4:
β^-, α, β^+

Correct Answer:
β^+, α, β^-

Solution:
${}^{A}_{Z}X \rightarrow \quad B \rightarrow \quad C \rightarrow \quad D$

$\quad\quad Z-1 \quad Z-3 \quad Z-2$

With β^+ decay ,Z decrease by 1 with α-decay,Z decrease by 2(from z-1 to z-3)

with β^- decay , Z increase by 1(from z-3 to z-2)

decay particles in the sequence are β^+, α, β^-

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Q. 31 Consider the following statements (A) and (B) and identify the correct answer:

(A) A zener dipole is connected in reverse bias when used as a voltage regulator

(B) The potential barrier of the p-n junction lies between 0.1 V to 0.3 V

Option 1:
(A) and (B) are are correct

Option 2:
(A) and (B) both are incorrect

Option 3:
(A) is correct and (B) is incorrect .

Option 4:
(A) is incorrect but (B) is correct.

Correct Answer:
(A) is correct and (B) is incorrect .

Solution:
The Zener dipole is always connected in reverse based on the potential barrier of the p-n junction lies between 0.1 to 0.3 V nearly.

Hence, the answer is the option (3).

Q. 32 A capacitor of capacitance 'C' is connected across an AC source of voltage V given by $V = V_o \sin \omega t$. The displacement current between the plates of the capacitor would then be given by:

Option 1:
$I_d = V_0 \omega C \cos \omega t$

Option 2:
$I_d = \frac{V_0}{\omega C} \cos \omega t$

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Option 3:

$I_d = \frac{V_0}{\omega C} \sin \omega t$

Option 4:

$I_d = V_0 \omega C \sin \omega t$

Correct Answer:

$I_d = V_0 \omega C \sin \omega t$

Solution:

$V = V_o \sin \omega t$

Displacement current is in magnitude equal to conduction current.

$$I_d = I_c = \frac{V}{X_C} = \frac{V_0 \sin \omega_t}{(1/\omega C)}$$

$$I_d = V_0 \omega C (\sin \omega t)$$

Hence, the answer is the option (4).

Q. 33 If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of:

Option 1:

$[M^2]\ [L^{-1}]\ [T^0]$

Option 2:

$[M]\ [L^{-1}]\ [T^{-1}]$

Option 3:

$[M]\ [L^0]\ [T^0]$

Option 4:

$[M^2]\ [L^{-2}]\ [T^{-1}]$

Correct Answer:

$[M^2]\ [L^{-1}]\ [T^0]$

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Solution:

$$[E] = [M^1\ L^2\ T^{-2}]$$

$$[G] = \left[\frac{Fr^2}{m_1 m_2}\right] = [M^{-1}L^3T^{-2})$$

$$\left[\frac{E}{G}\right] = [M^2L^{-1}T^0]$$

Hence, the answer is the option (1).

Q. 34 A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. if a parallel beam of light falling on 'A' leaves'B' as a parallel beam ,then the distance 'd' in cm will be:

Option 1:
25

Option 2:
15

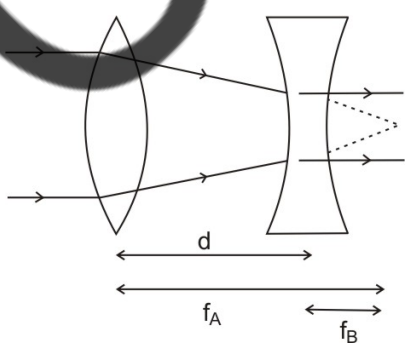
Option 3:
50

Option 4:
30

Correct Answer:
15

Solution:

$$f_A = 20\ \text{cm} \quad f_B = 5\ \text{cm}$$



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$f_A - f_B = d$

$d = 15 \text{ cm}$

Q. 35 If force [F], acceleration [A], and time [T] are chosen as the fundamental physical quantities, find the dimensions of energy.

Option 1:
$[F]\,[A]\,[T]$

Option 2:
$[F]\,[A]\,[T^2]$

Option 3:
$[F]\,[A]\,[T^{-1}]$

Option 4:
$[F]\,[A^{-1}]\,[T]$

Correct Answer:
$[F]\,[A]\,[T^2]$

Solution:

$[Energy] = [F]^x[A]^y[T]^Z$

$[M^1L^2T^{-2}] = [M^1L^1T^{-2}]^x[L^1T^{-2}]^y[T^1]^z$

$[M^1L^2T^{-2}] = M^xL^{x+y}T^{-2y-2x+z}$

$x = 1$

$x + y = 2 \quad ; y = 1$

$-2x - 2y + z = -2$

$z = -2 + 4 = 2$

$[\text{Energy}] = [F]^1[A]^1[T]^2$

Hence, the answer is the option (2).

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Q. 36 A Step-Down Transformer connected to an AC mains supply of 220V is made to operate at 11 V,44W lamp. Ignoring power losses in the Transformer, what is current in the primary circuit?

Option 1:
0.2A

Option 2:
0.4A

Option 3:
2A

Option 4:
4A

Correct Answer:
0.2A

Solution:
Since power loss is ignored

$$\eta = 1 = \frac{(power)\,output}{(power)\,input}$$

$$P_{input} = P_{output}$$

$$V_p I_p = V_s I_s$$

$$I_p = \frac{44}{220} = \frac{1}{5} = 0.2A$$

Hence, the answer is the option (1).

Q. 37 A ball of mas 0.15kg is dropped from a height 10m. strikes to the ground and rebounded to the same height.The magnitude of impluse imparted to the ball is (g=10m/s^2) nearly:

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Option 1:
0 kg m/s

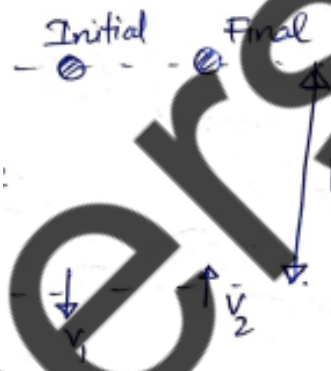
Option 2:
4.2 kg m/s

Option 3:
2.1 kg m/s

Option 4:
1.4 kg m/s

Correct Answer:
4.2 kg m/s

Solution:



$m = 0.15kg$

$h = 10m$

$\sin\ a$ ball rise to same height

$V_1 = V_2 = \sqrt{2gh}$

$= \sqrt{2 \times 10 \times 10}$

$V_1 = V_2 = 10\sqrt{2}$

Impulse=$\Delta P = P_f - P_i$

$= (+mv_2) - (-mv_1)$

$= (0.15)\,(v_2 + v_1)$

$= (0.15)\,20\sqrt{2} = 4.2kg\frac{m}{s}$

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Q. 38 A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection ,θ,is then given by :

Option 1:

$$\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Option 2:

$$\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{\frac{1}{2}}$$

Option 3:

$$\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{\frac{1}{2}}$$

Option 4:

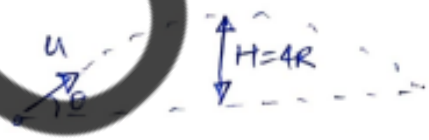
$$\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Correct Answer:

$$\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{\frac{1}{2}}$$

Solution:

$$T = \frac{2\pi R}{V}; V = \frac{2\pi R}{T}$$



$$H = 4R = \frac{u^2 \sin^2\theta}{2g}$$

$$u = v$$

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$$H = \frac{V^2 \sin^2\theta}{2g}$$

$$\sin^2 = \frac{2gh}{v^2} = \frac{2gH}{4\pi^2 R^2 \frac{}{T^2}}$$

$$\sin\theta = \left(\frac{2gHT^2}{4\pi^2 R^2}\right)^{\frac{1}{2}} = \left(\frac{2g\,(4g)\,T^2}{4\pi^2 R^2}\right)\frac{1}{2}$$

$$\sin\theta = \left(\frac{2gT^2}{\pi^2 R}\right)\frac{1}{2}$$

Q. 39 A particle of mass 'm' is projected with a velocity v=$Kv_e\,(K<1)$from the surface of the earth.$(v_e = escape - velocity)$. The maximum height above the surface reached by the particle is :

Option 1:

$$R\left(\frac{K}{1-K}\right)^2$$

Option 2:

$$R\left(\frac{K}{1+K}\right)^2$$

Option 3:

$$\frac{R^2K}{1+K}$$

Option 4:

$$\frac{RK^2}{1-K^2}$$

Correct Answer:

$$\frac{RK^2}{1-K^2}$$

Solution:

$$v = Kv_e\,(K<1)$$

$$TE_1 = TE_2$$

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$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{R}\right) = 0 + \left(\frac{-GMm}{R+h}\right)$$

$$\frac{1}{2}mxK^2 \times \frac{2GM}{R} - \frac{GMM}{R} = \frac{-GMm}{R+h}$$

$$\frac{GMm}{R}\left[K^2 - 1\right] = \frac{-GMm}{R+h}$$

$$\left[1 - K^2\right] = \frac{R}{R+h}$$

$$\frac{1}{(1-K^2)} = 1 + \frac{h}{R}$$

$$\frac{K^2R}{(1-K^2)} = h$$

Hence, the answer is the option (4).

Q. 40 A car starts from rest and accelerates at 5 m/s^2. At t=4s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t=6s?

(Take $g = 10m/s^2$)

Option 1:
20 m/s ,5m/s^2

Option 2:
20 m/s,0

Option 3:
$20\sqrt{2}m/s, 0$

Option 4:
$20\sqrt{2}m/s, 10m/s^2$

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Correct Answer:

$20\sqrt{2}m/s, 10m/s^2$

Solution:

$u = 0, a = 5m/s^2$[along horizontal direction]

at$t = 4s, v_x = u_x + a_x t$

$$v_x = 20m/s$$

The ball when released at $t = 4s$

From $t = 4s$ to $t = 6s$

$v_y = 0; v_y = u_y + a_y t$

$$v_y = 0 + (10)(2) = 20m/s$$

$$a = \sqrt{a_x^2 + a_y^2}$$

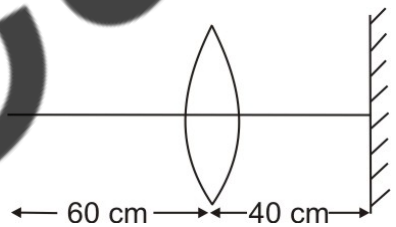
$$= \sqrt{0 + (10)^2} = 10m/s^2$$

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{(20)^2 + (20)^2}$$

$$v = 20\sqrt{2}m/s$$

Hence, the answer is the option (4).

Q. 41 A point object is placed at a distance of 60 cm from a convex lens of focal lenght 30 cm . If a plane mirror were put perpendicular to the principle axis of the lens and at a distance of 40cm form it, the final image would be formed at the distance of :



Option 1:

20 cm form the lens,it would be real image.

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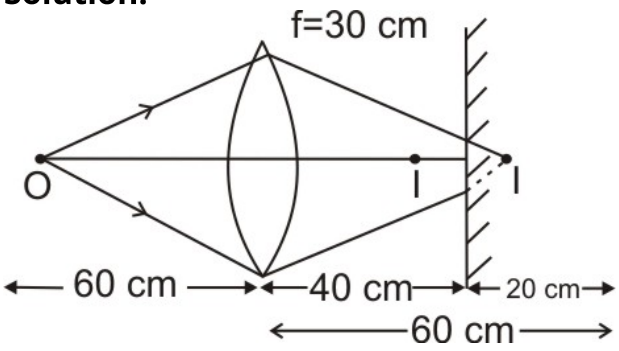
Option 2:
30 cm from the lens, it would be a real image .

Option 3:
30 cm from the plane mirror, it would be a virtual image .

Option 4:
20 cm from the plane mirror, it would be a virtual image .

Correct Answer:
20 cm form the lens,it would be real image.

Solution:



Object at 2f ,image at 2f

$I \rightarrow$ virtual object for plane mirror

$I' \rightarrow$ Real Image by plane mirror

Q. 42 Twenty seven drops of same size are charged at 220 V each.They combine to form a bigger drop.calculate the potential of the bigger drop.

Option 1:
660V

Option 2:
1320V

Option 3:
1520V

Option 4:
1980V

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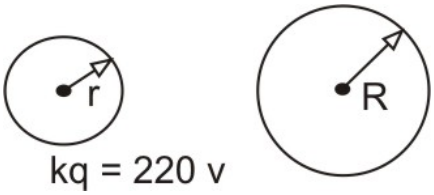
Correct Answer:
1980V

Solution:



$$V = \frac{kg}{r} = 220v$$

$$27 \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi R^3$$

$$R = 3r$$

Q Total $27q$(by Change convervation)

$$V_{Bigger\ drop} = \frac{kQ_{total}}{R} = \frac{27kq}{3r}$$

$$= q\,(220)$$

$$V_{Bigger\ drop} = 1980v$$

Q. 43 In the product $\overrightarrow{F} = q\left(\overrightarrow{v} \times \overrightarrow{B}\right) = q\,\overrightarrow{v} \times \left(B\hat{i} + B\hat{j} + B_0\hat{k}\right)$.

For $q = 1 and\ \overrightarrow{v} = 2\hat{i} + 4\hat{j} + 6\hat{k}\ \overrightarrow{F} = 4\hat{i} - 20\hat{j} + 12\hat{k}$. What will be the complete expression for $\overrightarrow{B}$?

Option 1:
$-8\hat{i} - 8\hat{j} - 6\hat{k}$

Option 2:
$-6\hat{i} - 6\hat{j} - 8\hat{k}$

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Option 3:

$8\hat{i}+8\hat{j}-6\hat{k}$

Option 4:

$6\hat{i}+6\hat{j}-8\hat{k}$

Correct Answer:

$-6\hat{i}-6\hat{j}-8\hat{k}$

Solution:

$$\bar{F}=q\left(\bar{v}\times\bar{B}\right)=q\bar{v}\times\left(B\hat{i}+B\hat{j}+B_0\hat{k}\right)$$

$$\left[4\hat{i}-20\hat{j}+12\hat{k}\right]=\left[2\hat{i}+4\hat{j}+6\hat{k}\right]\times\left[B\hat{i}+B\hat{j}+B_0\hat{k}\right]$$

$$=\hat{i}\left(4B_0-6B\right)-\hat{j}\left(2B_0-6B\right)+\hat{k}\left(-2B\right)$$

by equating the coefficients of equation ,we get

$-2B=12$

$B=-6$

$4B_0-6B=4$

$2B_0-6B=20$

$2B_0=-16, B_0=-8$

$B=-6$

$\bar{B}=B\hat{i}+B\hat{j}+B_0\hat{K}$

$\bar{B}=-6\hat{i}-6\hat{j}-8\hat{k}$

Hence, the answer is the option (2).

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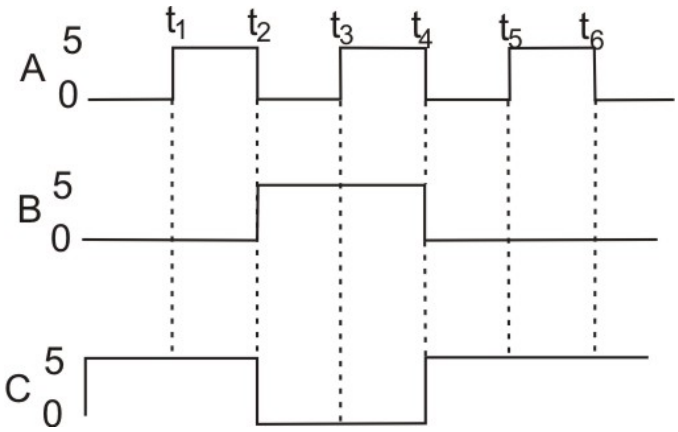
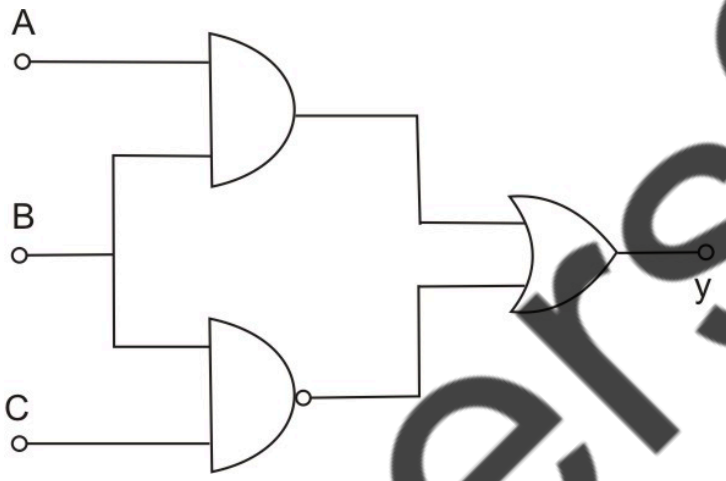
Q. 44 For the given circuit the input digital signals are applied at the terminals A,B and C ,What would be the output at the terminal y?





Option 1:

Option 2:

Option 3:

Option 4:

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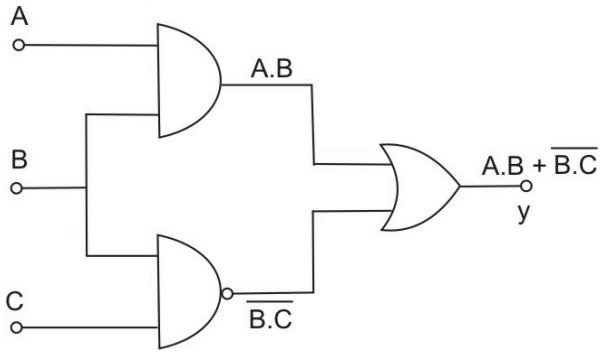
Correct Answer:

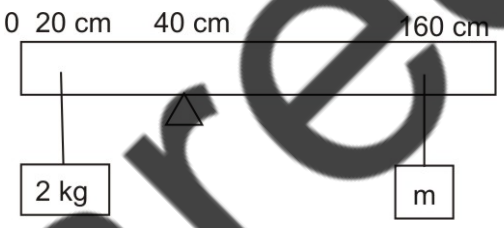
5 V

Solution:



A	B	C	A.B + $\overline{B.C}$ (Y)
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Q. 45 A uniform rod of lenght 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark.A mass of 2 kg is suspended form the rod at 20 cm and another unkonwn mass 'm' is suspended from the rod at 160 cm mark as shown in the figure.Find the value of 'm' such that the rod is in equilibrium (g=10m/s^2)



Option 1:

$\frac{1}{2}kg$

Option 2:

$\frac{1}{3}kg$

Option 3:

$\frac{1}{6}kg$

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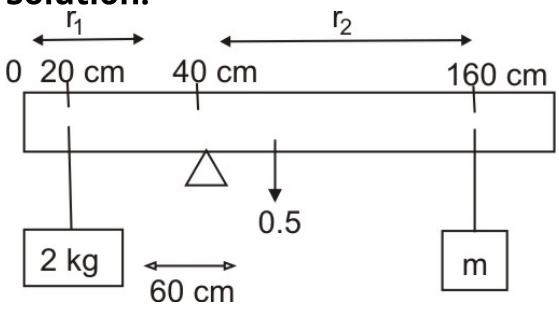
Option 4:

$\frac{1}{12}kg$

Correct Answer:

$\frac{1}{12}kg$

Solution:



for rotational equilibrium

$T_1 = T_2$

$(2 \times 10)\, r_1 = (m \times 10)\, r_2 + (0.5 \times 10)(0.6)$

$r_1 = 0.2m \quad r_2 = 1.2m$

$2 \times 0.2 - 0.3 = m\,(1.2)$

$\left(m = \frac{1}{12}kg\right)$

Q. 46 A uniform conducting wire of length 12a and resiatance 'R' is wound up as a current carrying coil in the shape of

(i) an equilateral triangle of side 'a'

(ii) a square of side 'a'

the magnetic dipole moments of the coil in each case respectively are

Option 1:

$\sqrt{3}Ia^2 and 3Ia^2$

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Option 2:

$3Ia^2 and Ia^2$

Option 3:

$3Ia^2 and 4Ia^2$

Option 4:

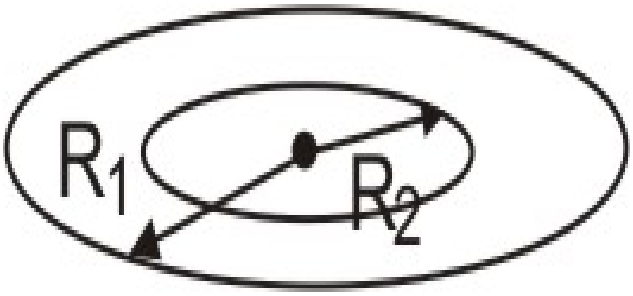
$4Ia^2 and 3Ia^2$

Correct Answer:

$\sqrt{3}Ia^2 and 3Ia^2$

Solution:

$Length = 12a$



$M_1 = n_1 I A_1$ $\qquad$ $M_2 = n_2 I A_2$

$4 \times I\left(\frac{1}{2} \times a \times \frac{\sqrt{3}}{2}a\right)$ $\qquad$ $= 3I\left(a^2\right)$

$M_1 = \sqrt{3}Ia^2$ $\qquad$ $M_2 = 3Ia^2$

Q. 47 Two conducting circular loops od raddi R_1 and R_2 are placed in the same plane with their centres coinciding .If $R_1 >> R_2$, the mutual inductance M between them will be directly proportional to :

Option 1:

$\frac{R_1}{R_2}$

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Option 2:

$\frac{R_2}{R_1}$

Option 3:

$\frac{R_1^2}{R_2}$

Option 4:

$\frac{R_2^2}{R_2}$

Correct Answer:

$\frac{R_2^2}{R_2}$

Solution:

$\phi = MI$

$BA = MI$

$\frac{\mu_0 I}{2R_1} \times TIR_2^2 = MI$

$M \propto \frac{R_2^2}{R_1}$

Q. 48 From a circular ring of mass 'M' and radius 'R' an are corresponding to a 90^0 sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the center of the ring and perpendicular to the plane of the ring is 'K'times MR^2, Then the value of 'K' is :

Option 1:

$\frac{3}{4}$

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Option 2:

$\frac{7}{8}$

Option 3:

$\frac{1}{4}$

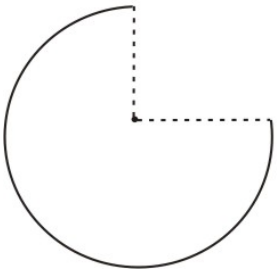
Option 4:

$\frac{1}{8}$

Correct Answer:

$\frac{3}{4}$

Solution:

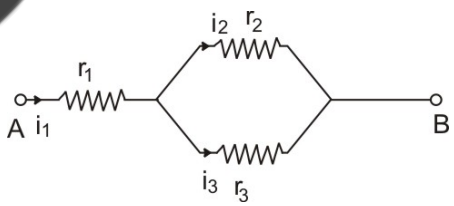


Remaining mass$= M' = \frac{3}{4}M$

$I = M'R^2 = \frac{3}{4}MR^2 = KMR^2$

$K = \frac{3}{4}$

Q. 49 Three resistance having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terme of resistances used in the circuit is :



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Option 1:

$$\frac{r_1}{r_2+r_3}$$

Option 2:

$$\frac{r_2}{r_2+r_3}$$

Option 3:

$$\frac{r_1}{r_1+r_2}$$

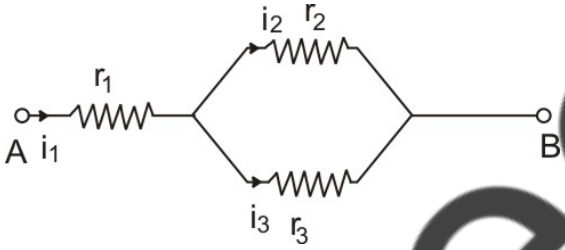
Option 4:

$$\frac{r_2}{r_1+r_3}$$

Correct Answer:

$$\frac{r_2}{r_2+r_3}$$

Solution:



$i_2 r_2 = i_3 r_3$ [resistance in parallel]

$i_2 + i_3 = i_1$[kirchoff's curret law]

$(i_1 - i_3)\, r_2 = i_3 r_3$

$i_1 r_2 = i_3 (r_3 + r_2)$

$$\frac{i_3}{i_1} = \frac{r_2}{r_2+r_3}$$

Q. 50 A series LCR circuit containing 5.0 H inductor $80\mu F$ capacitor and 40Ω resistor is connected to 230 V variable frequency AC source. The angular frequencies of the source at which power transferred to the circuit are half the power at the resonant angular frequency are likely to be :

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Option 1:
25 rad/s and 75 rad/s

Option 2:
50 rad/s and 25 rad/s

Option 3:
46 rad/s and 54 rad/s

Option 4:
42 rad/s and 58 rad/s

Correct Answer:
50 rad/s and 25 rad/s

Solution:

$L = 5H$

$C = 80\mu F$

$R = 40\Omega$

$P = \frac{1}{2}P_0$

$P_0 \rightarrow$ Power at resonance
$w_0 \rightarrow$ Resonance angular frequency
at resonance, $X_c = X_L$

$$w_0 C = \frac{1}{w_0 L}$$

$$w_0 = \frac{1}{\sqrt{LC}} = \frac{1}{20 \times 10^{-3}}$$

$$w_0 = 50\text{rad/s}$$

$P = V_{rms} I_{rms} \cos\phi_1$

$P_0 = V_{rms} I_{rms} \cos\phi_0$

$\cos\phi_0 = \frac{1}{2}$

$\cos\phi_1 = \frac{1}{2}$ $\cos\phi_o = \frac{R}{Z}$

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$$\frac{R}{Z_1} = \frac{1}{2} \qquad \phi_o = 0^o; \cos\phi_o = 1$$

$$80 = Z_1 = \sqrt{(X_L - X_C)^2 + R^2}$$

$$6400 = (X_L - X_C)^2 + 1600$$

$$4800 = (X_L - X_C)^2$$

$$4800 = \left(wL - \frac{1}{wC}\right)^2$$

$$40\sqrt{3} = \left(wL - \frac{1}{wC}\right)$$

$$w = 25 rad/s$$

$$w_o = 50\, rad/s \quad w = 25\ rad/s$$

Hence, the answer is the option (2).

Chemistry

Q. 1 Noble gases are named because of their inertness towards reactivity. Identify an incorrect statement about them.

Option 1:
Noble gases are sparingly soluble in water.

Option 2:
Noble gases have very high melting and boiling point.

Option 3:
Noble gases have weak dispersion force.

Option 4:
Noble gases have large positive values of electron gain enthalpy.

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Correct Answer:
Noble gases have very high melting and boiling point.

Solution:
Noble gases are usually monoatomic and have very weak interaction forces. They usually have very low melting and boiling points.
The stronger the interaction between particles of gas more boiling and melting points.

Hence, the answer is the option (2).

Q. 2 The correct option for the number of body centred unit cells in all 14 types of Bravais lattice unit cells is:

Option 1:
7

Option 2:
5

Option 3:
2

Option 4:
3

Correct Answer:
3

Solution:
Body-centred unit cells are possible for Orthorhombic, cubic and tetragonal types of Bravais lattice unit cells.

Hence, out of the 14 possible bravais lattice, body-centred unit cells exist for only 3.

Therefore, the correct option is (4).

Q. 3 Which are among the following is the correct option for right relationship between C_p and C_v for one mole of ideal gas?

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Option 1:
$C_p + C_v = R$

Option 2:
$C_p - C_v = R$

Option 3:
$C_p = RC_v$

Option 4:
$C_v = RC_p$

Correct Answer:
$C_p - C_v = R$

Solution:
The relation between $C_p \ and \ C_v$ for one mole of an ideal gas is given as

$$C_P - C_V = R.$$

Hence, the answer is the option (3).

Q. 4 Zr(Z=40) and Hf(Z=72) have similar atomic and ionic raddi because of:

Option 1:
Belonging to same group

Option 2:
Diagonal relationship

Option 3:
Lanthanoid contraction

Option 4:
Having Similar chemical properties

Correct Answer:
Lanthanoid contraction

Solution:

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Zr and Hf have an almost similar size. This is because of Lanthanoid contraction.

Hence, the answer is the option (3).

Q. 5 The Maximum Temperature that can be achieved in a blast furnace is :

Option 1:
Upto 1200 K

Option 2:
Upto 2200 K

Option 3:
Upto 1900 K

Option 4:
Upto 5000 K

Correct Answer:
Upto 2200 K

Solution:
The maximum temperature that can be achieved in a Blast furnace is 2200K.

The correct answer is option 2.

Q. 6 Statement I: Acid strength increases in the order given as
$HF << HCl << HBr << HI.$

Statement II: As the size of elements F, Cl, Br, I increases down the group, the bond strength of $HF, HCl, HBr, and\ HI$ decreases and so the acid strength increases

In the light of the above statements, choose the correct answer from the option given below

Option 1:
Both Statement I and Statement II are true

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Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is correct but Statement II is false.

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Both Statement I and Statement II are true

Solution:
Statement I and Statement II are both correct.

As the size of the halogen increases, the acidic strength of the hydrogen halide increases.

Also, as the difference in the size of the halogen and hydrogen increases, the bond length increases and the bond dissociation energy decreases.

Lower is bond dissociation energy, more is acidic strength.
The correct answer is option 1.

Q. 7 The compound which shows metamerism is :

Option 1:
C_5H_{12}

Option 2:
C_3H_8O

Option 3:
C_3H_6O

Option 4:
$C_4H_{10}O$

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Correct Answer:

$C_4H_{10}O$

Solution:

Metamerism is shown by compounds that can have different alkylation around a central atom or a group.

$C_4H_{10}O$ can show metamerism which can be illustrated as

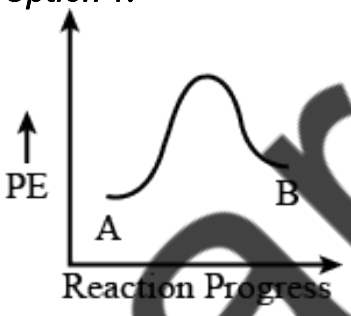
$C_2H_5 - O - C_2H_5 \; ; \; CH_3 - O - C_3H_7$

Here the alkylation is different in the compounds around the heteroatom O.

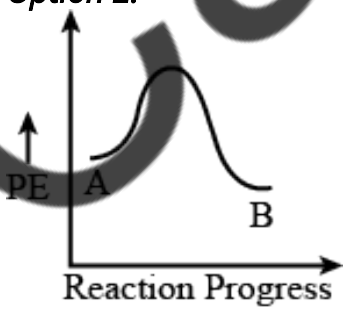
Hence, the answer is the option (4).

Q. 8 For a reaction $A \rightarrow B$, the enthalpy of reaction is $-4.2kJmol^{-1}$and enthalpy of activation is 9.6 kJ mol^{-1} The correct potential energy profile for the reaction is shown in the option.

Option 1:



Option 2:



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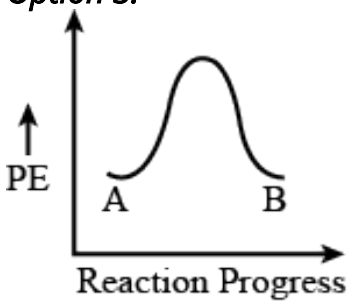
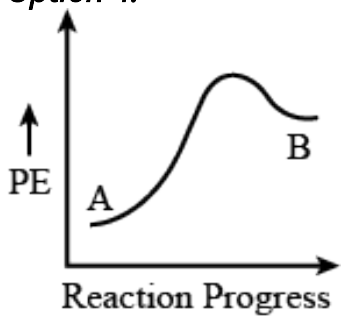
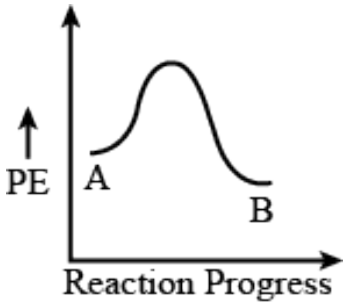
Option 3:



Option 4:

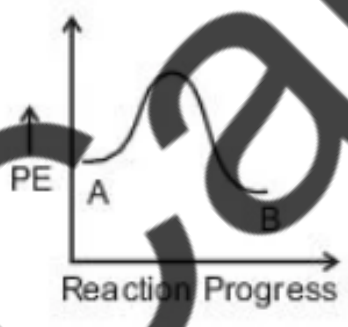


Correct Answer:



Solution:

For an exothermic reaction, the enthalpy of the reactant will be greater than the enthalpy of the product and as a result, the energy profile diagram will look like



The correct answer is option 2.

Q. 9 The right option for the statement "Tyndall effect is exhibited by" is :

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Option 1:
NaCl Solution

Option 2:
Glucose solution

Option 3:
Starch Solution

Option 4:
Urea Solution

Correct Answer:
Starch Solution

Solution:
Tyndall effect is shown by Colloidal solutions.

Out of the given solutions, only starch solution can have particles of colloidal size.

Urea, glucose, and NaCl form true solutions and hence, cannot show Tyndall Effect.

The correct answer is option 3.

Q. 10 BF_3 is a planar and electron-deficient compound. Hybridization and number of electrons around the central atom, respectively are:

Option 1:
$sp^3 \ and \ 4$

Option 2:
$sp^3 \ and \ 6$

Option 3:
$sp^2 \ and \ 6$

Option 4:
$sp^2 \ and \ 8$

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Correct Answer:
$sp^2\ and\ 6$

Solution:
BF_3 is a trigonal planar molecule with 3 sigma bonds formed by Boron with the 3 Fluorine atoms.

$$H = \frac{3+3}{2} = 3$$

Hence, the hybridization is sp^2 and the number of electrons around boron is 6.

Hence, the answer is the option (3).

Q. 11 Match List -I with List -II

List I	List II
(a) Pcl_5	(i) Square pyramidal
(b) SF_6	(ii) Trigonal Planar
(c) BrF_5	(iii) Octahedral
(d) BF_3	(iv) Trigonal bipyramidal

Choose the corret Answer from the given below

Option 1:
(a)-(iv),(b)-(iii),(c)-(i) ,(d)-(ii)

Option 2:
(a)-(ii),(b)-(iii),(c)-(iv),(d)-(i)

Option 3:
(a)-(iii),(b)-(i),(c)-(iv),(d)-(ii)

Option 4:
(a)-(iv),(b)-(iii),(c)-(ii),(d)-(i)

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Correct Answer:
(a)-(iv),(b)-(iii),(c)-(i) ,(d)-(ii)

Solution:
To calculate hybridization

$$H = \frac{V + M - Q}{2}$$

Where V is the number of valence electrons in the central atom

M is the Number of surrounding atoms

Q is the charge present on the molecule (if any)
PCl_5 has trigonal bipyramidal shape and sp^3d hybridization.

$$H = \frac{5+5}{2} = 5 \quad sp^3d$$

SF_6 has sp^3d^2 hybridization and octahedral geometry.

$$H = \frac{6+6}{2} = 6 \quad sp^3d^2$$

BrF_5 has sp^3d^2 hybridization and square pyramidal geometry.

$$H = \frac{7+5}{2} = 6 \quad sp^3d^2$$

BF_3 has sp^2 hybridization and trigonal planar geometry.

$$H = \frac{3+3}{2} = 3 \quad sp^2$$

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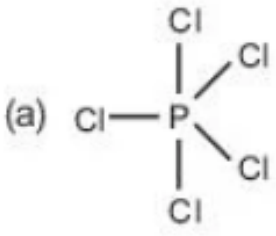
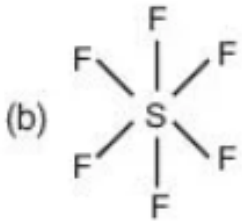
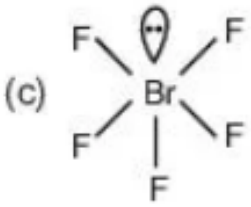


Trigonal Bipyramidal



Octahedral



Square Pyramidal

Trigonal Planar

Hence, the correct answer is option 1.

Q. 12 The structures of beryllium chloride in solid state and vapour phase, are

Option 1:
Chain and dimer, respectively

Option 2:
Linear in both

Option 3:
Dimer and Linear,respectively

Option 4:
Chain in both

Correct Answer:
Chain and dimer, respectively

Solution:
$BeCl_2$ exists as a polymeric solid and a dimeric gas. The structures in the solid state and the gaseous state is given below:

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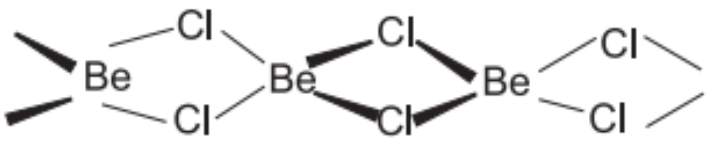
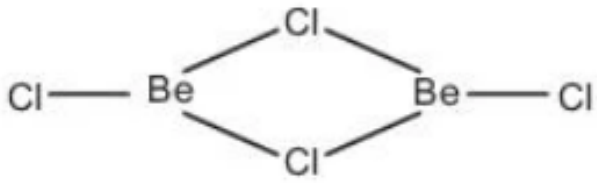


Structure of $BeCl_2$ Solid



Structure of $BeCl_2$ dimer in gaseous state

Hence, the correct answer is option 1.

Q. 13 The correct sequence of bond enthalpy of 'C-X' bond is :

Option 1:

$CH_3 - F < CH_3 - Cl < CH_3 - Br < CH_3 - I$

Option 2:

$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$

Option 3:

$CH_3 - F < CH_3 - Cl > CH_3 - Br > CH_3 - I$

Option 4:

$CH_3 - Cl > CH_3 - F > CH_3 - Br > CH_3 - I$

Correct Answer:

$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$

Solution:

Bond strength depends upon the extent of overlap. As the size difference increases, the bond energy or the bond strength decreases as the overlap becomes less effective.

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Thus, as the size of the halogen increases, the C-X bond energy decreases.

Therefore, the correct order is

$$CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$$

Hence, the answer is the option (2).

Q. 14 Which one of the following polymers is prepared by addition polymerisation?

Option 1:
Teflon

Option 2:
Nylon - 6,6

Option 3:
Novolac

Option 4:
Dacron

Correct Answer:
Teflon

Solution:
Teflon is Polytetrafluoroethene and it is prepared by the addition polymerisation of tetrafluoroethene.

Nylon, Novolac and Dacron are condensation polymers.

Hence, the correct answer is Option (1)

Q. 15 Dihedral angle of least stable conformer of ethane is :

Option 1:
120^0

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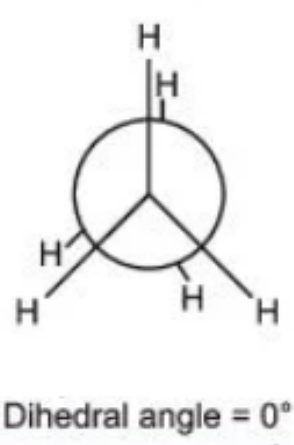
Option 2:
180^0

Option 3:
60^0

Option 4:
0^0

Correct Answer:
0^0

Solution:
The least stable conformer of ethane is the eclipsed form in which the dihedral angle is Zero.



Dihedral angle = 0°

Q. 16 An organic compound contains 78% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is :[Atomic wt.C is 12, H is 1]

Option 1:
CH

Option 2:
CH_2

Option 3:
CH_3

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Option 4:
CH_4

Correct Answer:
CH_3

Solution:
The mass ratio of Carbon : Hydrogen = 78 : 22

$\therefore$ the molar ratio of Carbon : Hydrogen = 6.5 : 22

$\therefore$ the simplest ratio of Carbon : Hydrogen = 1 : 3.38

Thus, based on the given data, the most probable answer is $\mathrm{CH_3}$

Hence, the answer is the option (3).

Q. 17 The molar conductance of NaCl ,HCl,and $\mathrm{CH_2COONa}$ at infinite dilution are 126.45, 426.16 and $91.0\ \mathrm{S\ cm^2 mol^{-1}}$ respectively.The molar conductance of $\mathrm{CH_3COOH}$ at infinte dilution is _____.

Choose the right option for your answer:

Option 1:
$201.28\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 2:
$390.71\ \mathrm{S\ cm^2\ mol^{-1}}$

Option 3:
$698.28\ \mathrm{S\ cm^2 mol^{-1}}$

Option 4:
$540.48\ \mathrm{S\ cm^2\ mol^{-1}}$

Correct Answer:
$390.71\ \mathrm{S\ cm^2\ mol^{-1}}$

Solution:
Given,

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$\Lambda_m^o, NaCl = 126.45$

$\Lambda_m^o, HCl = 426.16$

$\Lambda_m^o, CH_3COONa = 91$

$\Lambda_m^o, CH_3COOH = ?$

We know from Kohlrausch's Law of independent migration of ions,

$\Lambda_m^o, CH_3COOH = \Lambda_m^o, CH_3COONa + \Lambda_m^o, HCl - \Lambda_m^o, NaCl$

$\Lambda_m^o, CH_3COOH = 91 + 426.16 - 126.45 = 390.71$

Hence, the answer is the option (2).

Q. 18 Given below are two statements :

Statement I :

Aspirin and Paracetamol belong to the class of narcotic analgesics.

Statement II :

Morphine and Heroin are non-narcotic analgesics.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both **Statement I** and **Statement II** are true.

Option 2:
Both **Statement I** and **Statement II** are false.

Option 3:
Statement I is correct but **Statement II** is false.

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Option 4:
Statement I is incorrect but **Statement II** is true.

Correct Answer:
Both **Statement I** and **Statement II** are false.

Solution:
Aspirin and Paracetamol are Non Narcotic analgesics while Morphine and Heroin are Narcotic Analgesics.

Thus, Statement I and II are both incorrect.

Q. 19 The following solutions were prepared by dissolving 10 g of glucose $(C_6H_{12}O_6)$ in 250 ml of water (P_1), 10 g of urea (CH_4N_2O) in 250 ml of water (P_2), and 10 g of sucrose $(C_{12}H_{22}O_{11})$ in 250 ml of water (P_3). The right option for the decreasing order of osmotic pressure of these solutions is :

Option 1:
$P_2 > P_1 > P_3$

Option 2:
$P_1 > P_2 > P_3$

Option 3:
$P_2 > P_3 > P_1$

Option 4:
$P_3 > P_1 > P_2$

Correct Answer:
$P_2 > P_1 > P_3$

Solution:
We know that,

$$\pi = CRT$$

So under similar conditions of temperature and volume of solution, the species with the greater number of moles has a greater value of Osmotic Pressure.

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Order of moles : Urea > Glucose > Sucrose

Order of Osmotic Pressure:

$$P_2 > P_1 > P_3$$

Hence, the answer is the option (1).

Q. 20 Among the following alkaline earth metal halides, one which is covalent and soluble in organic solvents is :

Option 1:
Calcium chloride

Option 2:
Strontium chloride

Option 3:
Magnesium chloride

Option 4:
Beryllium chloride

Correct Answer:
Beryllium chloride

Solution:
For chlorides of group 2 elements, going down the group covalent character increases. This is because going down the group charge to size ratio decreases and the polarizing power of cation decreases and the lower is polarising power, the more ionic character.

Polarizing power of group 2 elements is $Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$

and, the covalent character is

$$BeCl_2 > MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$$

hence, the correct answer is option 4.

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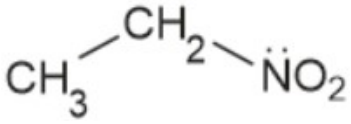
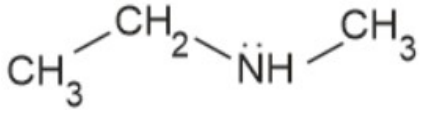
Q. 21 Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.

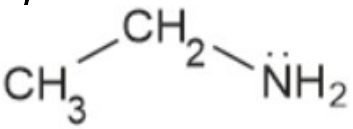
Option 1:



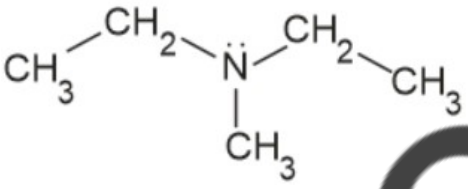
Option 2:



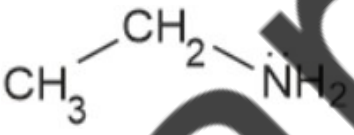
Option 3:



Option 4:



Correct Answer:



Solution:

Hinsberg's reagent test is given by both primary as well as secondary amines. However, it is the primary amines only that give a product which is soluble in Alkali.

This is because there is an acidic hydrogen ($N-H$) present in the addition product with Benzene Sulphonyl Chloride for a Primary amine.

Q. 22 Right option for the number of tetrahedral and octahedral voids in hexagonal primitive unit cell are :

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Option 1:
$8, 4$

Option 2:
$6, 12$

Option 3:
$2, 1$

Option 4:
$12, 6$

Correct Answer:
$12, 6$

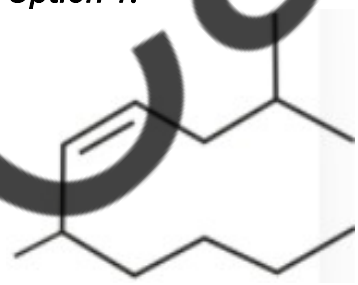
Solution:
In any closed-packed structure containing N number of particles, the number of Tetrahedral Voids is 2N and the number of Octahedral Voids is N.

Now since in a hexagonal closed packed structure, the number of particles is 6, the number of Tetrahedral and Octahedral Voids will be 12 and 6 respectively.

Therefore, the correct option is (4).

Q. 23 The correct structure of 2,6- Dimethyl-dec-4-ene is :

Option 1:



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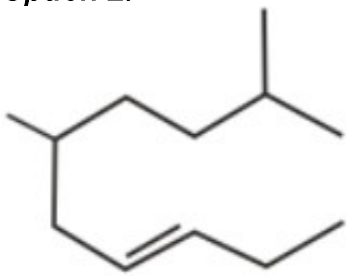
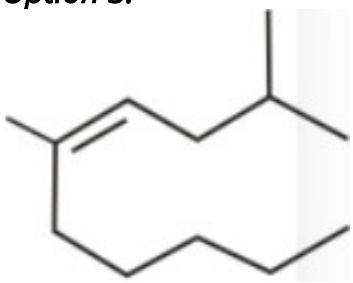
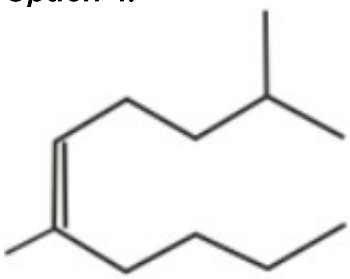
Option 2:



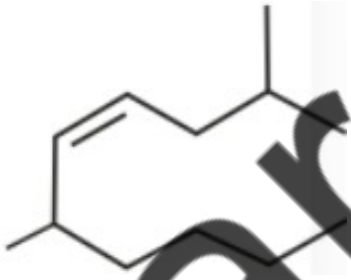
Option 3:



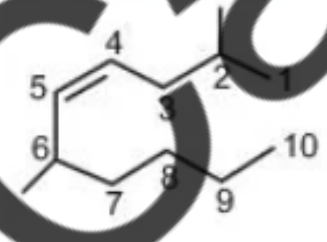
Option 4:



Correct Answer:



Solution:



2,6-Dimethyl-dec-4-ene

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Q. 24 The RBC deficiency is deficiency disease of :

Option 1:
Vitamin B_{12}

Option 2:
Vitamin B_6

Option 3:
Vitamin B_1

Option 4:
Vitamin B_2

Correct Answer:
Vitamin B_{12}

Solution:
Factual Question.

RBC deficiency is caused by deficiency of Vitamin B_{12}

Q. 25 Which one of the following methods can be used to obtain highly pure metal which is liquid at room temperature?

Option 1:
Electrolysis

Option 2:
Chromatography

Option 3:
Distillation

Option 4:
Zone refining

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Correct Answer:
Distillation

Solution:
Distillation is used as a process of refining metals having low boiling points like $Zn\ , Hg\ and\ Cd.$.

Hence, the correct answer is option 3.

Q. 26 What is the IUPAC name of the organic compound formed in the following chemical reaction ?

Acetone $\xrightarrow[(ii)H_2O,H^+]{(i)C_2H_5MgBr,\text{dry Ether}}$ product

Option 1:
2 methyl propan-2-ol

Option 2:
pentan - 2-ol

Option 3:
pantan-3-ol

Option 4:
2 - methyl butan- 2- ol

Correct Answer:
2 - methyl butan- 2- ol

Solution:

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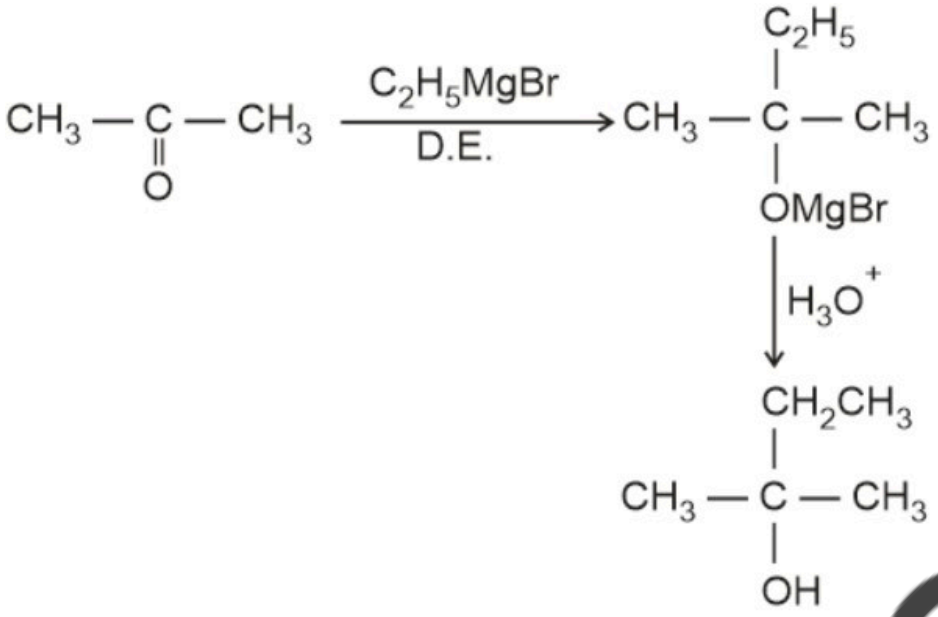
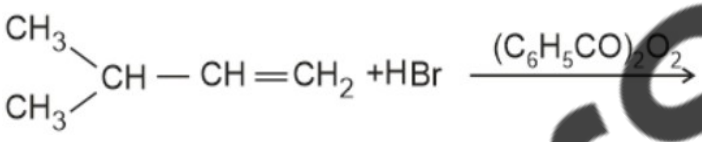
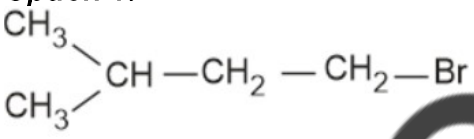


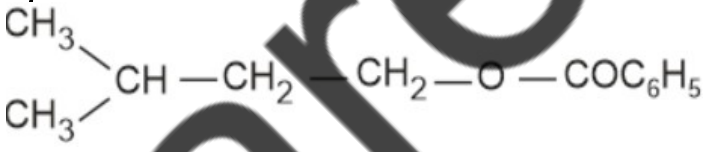
Q. 27 The major product of the following chemical reaction is :



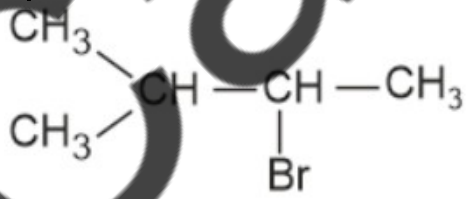
Option 1:



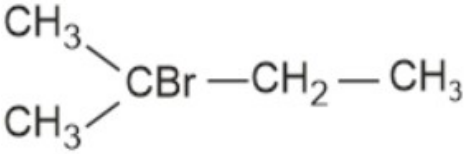
Option 2:



Option 3:



Option 4:



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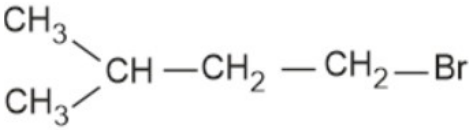
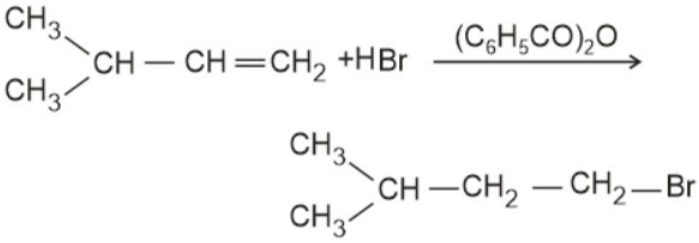
Correct Answer:



Solution:



according to antimarkovnikov Rules,

Q. 28 The **incorrect** statement among the following is :

Option 1:
Actionoid contraction is greater for element to element than Lanthanoid contraction.

Option 2:
Most of the trivalent Lanthanoid ions are colorless in the solid state.

Option 3:
Lanthanoids are good conductors of heat and electricity.

Option 4:
Actionoids are highly reactive metals, especially when finely divided.

Correct Answer:
Most of the trivalent Lanthanoid ions are colorless in the solid state.

Solution:
Most of the trivalent lanthanoids are Colored in both the Solid as well as the aqueous state.

Other statements are all correct.

Hence, the answer is the option (2).

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Q. 29 A particular station of All India Radio, New Dehli, broadcasts on a frequency of 1,368 kHz (kilohertz), the wavelength of the electromagnetic radiation emitted by the transmitter is : [speed of light, $\mathrm{e} = 3.0 \times 10^{8} \mathrm{ms}^{-1}$].

Option 1:
$219.3\ m$

Option 2:
$219.2\ m$

Option 3:
$2192\ m$

Option 4:
$21.92\ cm$

Correct Answer:
$219.3\ m$

Solution:
We know that,

$$\nu = \frac{c}{\lambda}$$

$$\therefore \lambda = \frac{\mathrm{c}}{\nu} = \frac{3 \times 10^{8}\ \mathrm{m/s}}{1368 \times 10^{3}\ \mathrm{s}} = 219.3\ \mathrm{m}$$

Hence, the answer is the option (1).

Q. 30 Which of the following reactions is the metal displacement reaction ? Choose the right option.

Option 1:
$2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

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Option 2:

$$Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2CrCl_3$$

Option 3:

$$Fe + 2HCl \rightarrow FeCl_2 + H_2\uparrow$$

Option 4:

$$2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2\uparrow$$

Correct Answer:

$$Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2CrCl_3$$

Solution:

Decomposition reaction leads to the breakdown of a compound into two or more components at least one of which must be in the elemental state.

Both reactions (1) and (4) are examples of decomposition reactions.

In a **displacement reaction**, an ion (or an atom) in a compound is replaced by an ion (or an atom) of another element. There are two types of displacement reactions:

Metal displacement - A metal in a compound can be displaced by another metal in the uncombined state

Non-metal displacement - The non-metal displacement redox reactions include hydrogen displacement and a rarely occurring reaction involving oxygen displacement.

Reactions (2) and (3), both are examples of displacement reactions,

But reaction (2) is an example of a metal displacement reaction because metal is displacing there.

Therefore, the correct option is (2).

Q. 31 Choose the correct option for graphical representation of Boyle's law, which shows a graph of pressure vs. volume of a gas at different temperatures:

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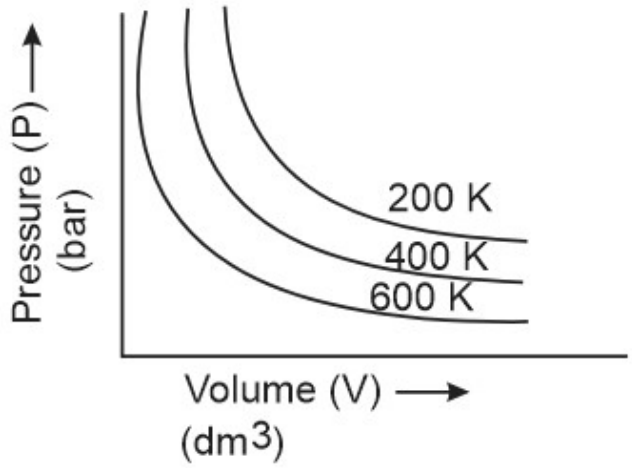
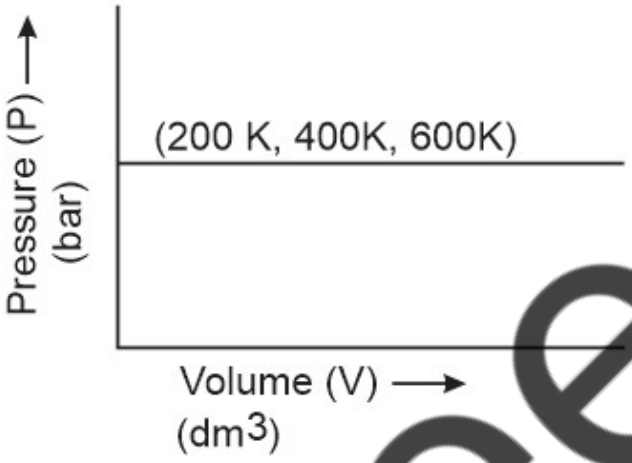
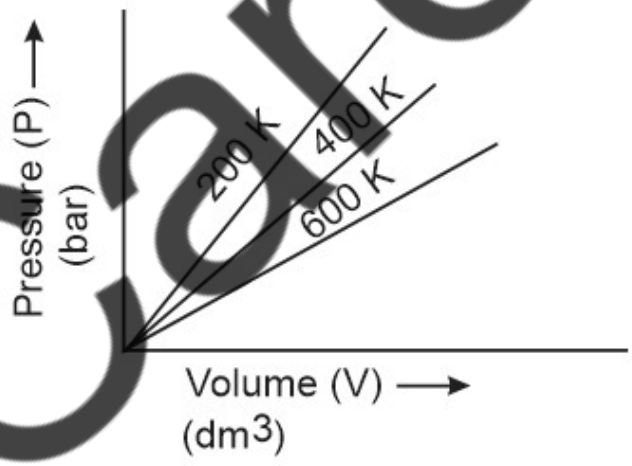
Option 1:



Option 2:



Option 3:



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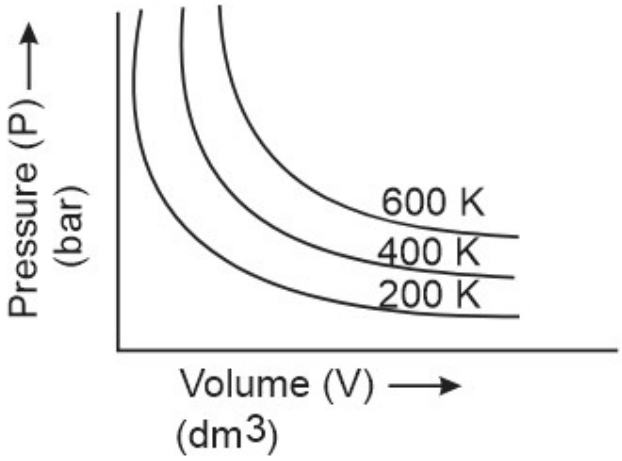
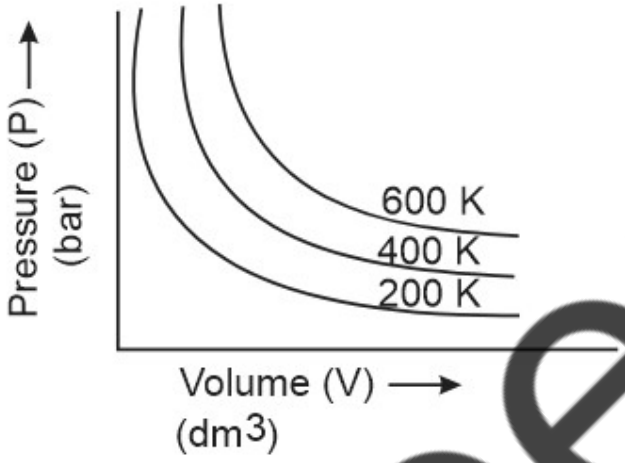
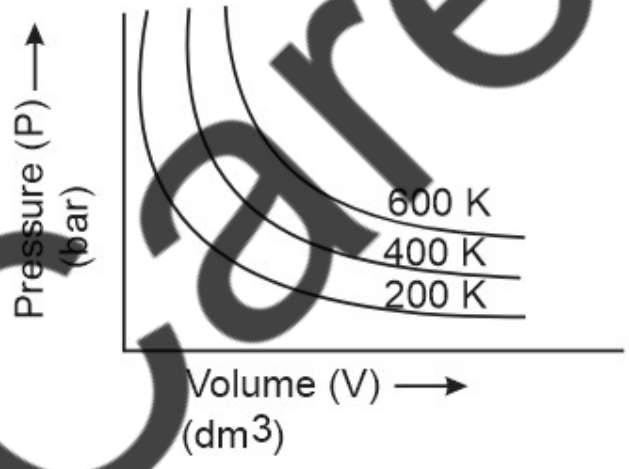
Option 4:



Correct Answer:



Solution:



Boyle's Law is -

at a constant temperature, the pressure of a fixed amount (i.e., number of moles n) of gas varies inversely with its volume.

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$$P \propto \frac{1}{V} \Rightarrow P = \frac{k}{V} \Rightarrow PV = k$$

Where k is proportionality constant and equal to nRT.

The graph between P vs. V should be a rectangular hyperbola.

If T increases the value of k increases then at the same value of V, P will increase due to k increases.

Therefore, the correct option is (4).

Q. 32 The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is :

Option 1:
8.50

Option 2:
5.50

Option 3:
7.75

Option 4:
6.25

Correct Answer:
7.75

Solution:
We know the formula for the pH of salt of the weak acid and weak base -

$$pH = 7 + \frac{1}{2}(pK_a - pK_b)$$

Given,

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$$pK_a = 3.27 \text{ and } pK_b = 4.77$$

So,

$$pH = 7 + \frac{1}{2}(4.77 - 3.27)$$

$$pH = 7.75$$

Hence, the answer is the option (3).

Q. 33 The major product formed in dehydrohalogenation reaction of 2 - Bromo pentane is Pent - 2-ene. This product formation is based on ?

Option 1:
Saytzeff's Rule

Option 2:
Hund's Rule

Option 3:
Hofmann Rule

Option 4:
Huckel's Rule

Correct Answer:
Saytzeff's Rule

Solution:
Saytzeff's Rules:

" More substituted alkene is the major product"

Hence, the answer is the option (1).

Q. 34 Ethylene diaminetetraacetate (EDTA) ion is :

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Option 1:
Hexadentate ligand with four "O" and two "N" donor atoms

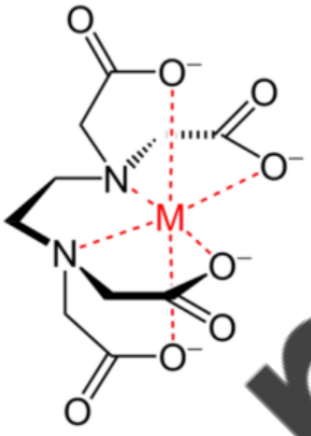
Option 2:
Unidentate ligand

Option 3:
Bidentate ligand with two "N" donor atoms

Option 4:
Tridentate ligand with three "N" donor atoms

Correct Answer:
Hexadentate ligand with four "O" and two "N" donor atoms

Solution:
The structure of EDTA is -



It is a Hexadentate ligand with four "O" and two "N" donor atoms.

Therefore, the correct option is (1).

Q. 35 Tritium, a radioactive isotope of hydrogen, emits which of the following particles?

Option 1:
$\text{Beta}\ (\beta^-)$

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Option 2:
Alpha (α)

Option 3:
Gamma (γ)

Option 4:
Neutron (n)

Correct Answer:
Beta (β^-)

Solution:
Tritium is a radioactive isotope of Hydrogen and it emits low-energy Beta particles (β^-).

Therefore, the correct option is (1).

Q. 36 Match List - I with List - II

List - I	List - II
(a) $[Fe(CN)_6]^{3-}$	(i) 5.92 BM
(b) $[Fe(H_2O)_6]^{3+}$	(ii) 0 BM
(c) $[Fe(CN)_6]^{4-}$	(iii) 4.90 BM
(d) $[Fe(H_2O)_6]^{2+}$	(iv) 1.73 BM

Choose the correct options from the given below:

Option 1:
(a) - (iv), (b) - (ii), (c)- (i), (d)- (iii)

Option 2:
(a) - (ii), (b) - (iv), (c)- (iii), (d)- (i)

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Option 3:
(a) - (i), (b) - (iii), (c)- (iv), (d)- (ii)

Option 4:
(a) - (iv), (b) - (i), (c)- (ii), (d)- (iii)

Correct Answer:
(a) - (iv), (b) - (i), (c)- (ii), (d)- (iii)

Solution:
CN^- is a strong field ligand and will cause pairing while H_2O is a weak field ligand and will not cause pairing

Thus, Fe^{3+} which has d^5 configuration and will have 1 unpaired electron in the complex with CN^- and 5 unpaired electrons in the complex with H_2O

And, Fe^{2+} which has d^6 configuration and will have 0 unpaired electron in the complex with CN^- and 4 unpaired electrons in the complex with H_2O

Complex	unpaired electrons (n)	Magnetic moment (BM)
$[Fe(CN)_6]^{3-}$	1	1.73
$[Fe(H_2O)_6]^{3+}$	5	5.92
$[Fe(CN)_6]^{4-}$	0	0
$[Fe(H_2O)_6]^{2+}$	4	4.9

Hence, the answer is the option (4).

Q. 37 In which one of the following arrangements the given sequence is not strictly according to the properties indicated against it ?

Option 1:
$HF < HCl < HBr < HI$: Increasing acidic strength

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Option 2:
$H_2O < H_2S < H_2Se < H_2Te$: Increasing pK_a values

Option 3:
$NH_3 < PH_3 < AsH_3 < SbH_3$: Increasing acidic character

Option 4:
$CO_2 < SiO_2 < SnO_2 < PbO_2$: Increasing oxidizing power

Correct Answer:
$H_2O < H_2S < H_2Se < H_2Te$: Increasing pK_a values

Solution:
(1) is correct. On moving down the group, acidic strength increases

(2) is incorrect. The greater the acidic strength, the greater the Ka value and the lesser the pKa value.

(3) is correct. On moving down the group, acidic strength increases

(4) is correct. On moving down group 14, the stability of the +4 oxidation state decreases and the tendency to accept electrons increases. Thus, the oxidizing power increases

Hence, the answer is the option (2).

Q. 38 Choose the correct option for the total pressure (in atm.) in a mixture of 4 g O_2 and 2 g H_2 confined in a total volume of one liter at 0^oC is :

[Given $R = 0.082L \text{ atm } \text{ mol}^{-1}K^{-1}, T = 273 \ K$]

Option 1:
2.518

Option 2:
2.602

Option 3:
25.18

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Option 4:
26.02

Correct Answer:
25.18

Solution:
In the question, the pressure needed to find, Volume, and Temperature are given.

We can use PV =nRT formula.

V = 1 litre

T = 0^oC = 273 K

n = mole = mass/molar mass

mass is given of each. and molar mass we know that.

so,

$$n_{O_2} = \frac{4}{32} = \frac{1}{8}$$

$$n_{H_2} = \frac{2}{2} = 1$$

So, total moles will be -

$$n_t = \frac{1}{8} + 1 = \frac{9}{8}$$

The formula will be for mixture -

$$P_tV = n_tRT$$

$$P_t \times 1 = \frac{9}{8} \times 0.082 \times 273$$

$$P_t = 25.18 \text{ atm}$$

Therefore, the correct option is (3).

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Q. 39 Which of the following molecules is non-polar in nature ?

Option 1:
$POCl_3$

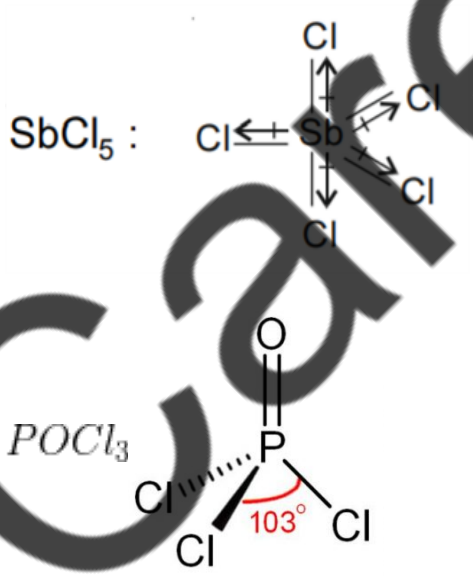
Option 2:
CH_2O

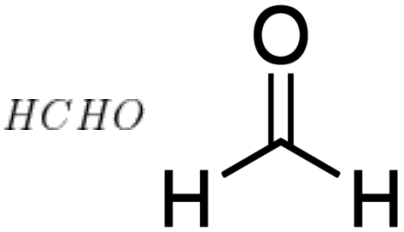
Option 3:
$SbCl_5$

Option 4:
NO_2

Correct Answer:
$SbCl_5$

Solution:
$SbCl_5$ molecule is non-polar in nature due to its symmetrical shape -





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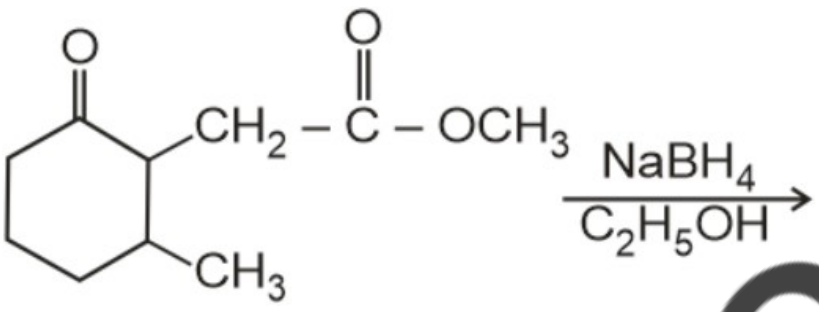
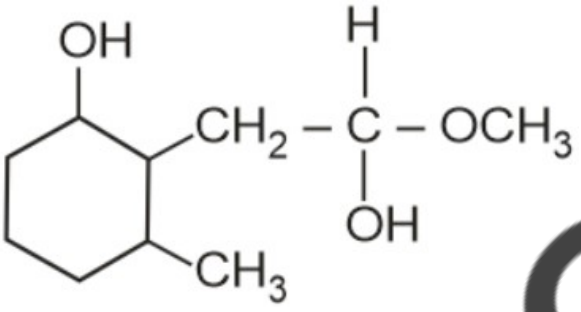
NO_2 has a bent shape and asymmetric hence is also polar in nature.,

Therefore, the correct option is (3).

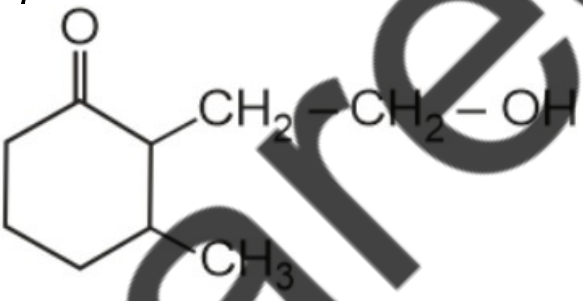
Q. 40 The product formed in the following chemical reaction is :



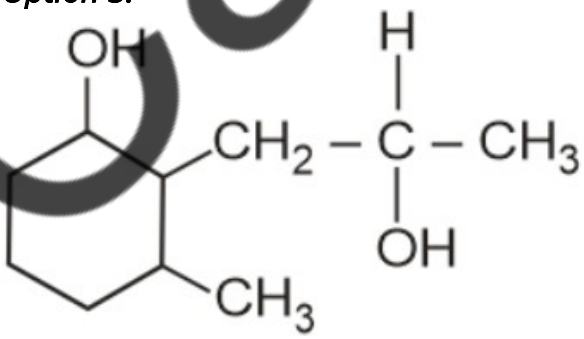
Option 1:



Option 2:



Option 3:



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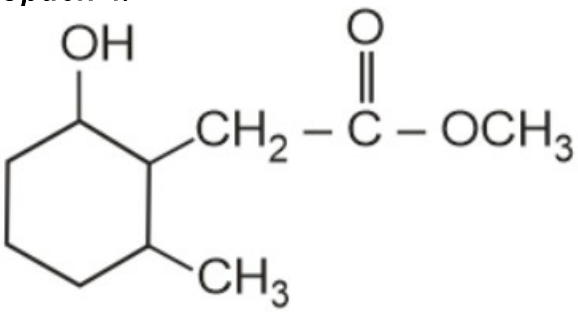
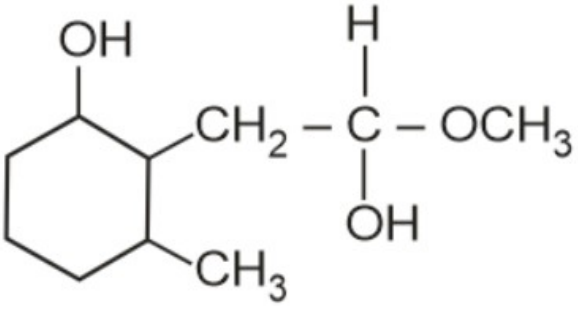
Option 4:



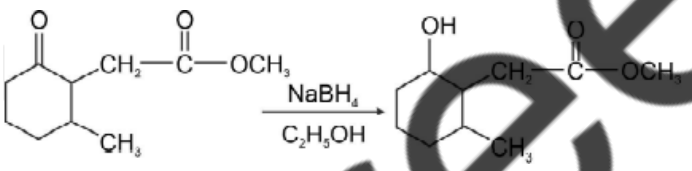
Correct Answer:



Solution:

$NaBH_4/C_2H_5OH$ is a weak reducing agent. It cannot reduce −COOR group

It reduces group to group



Q. 41 The slope of the Arrhenius Plot $ln\ k\ v/s\ \frac{1}{T}$ the first-order reaction is $-5 \times 10^3 K$. The value of E_a of the reaction is. Choose the correct option for your answer.

$\left[\text{Given R} = 8.314\ \text{JK}^{-1}\text{mol}^{-1}\right]$

Option 1:

$41.5\ \text{kJ mol}^{-1}$

Option 2:

$83.0\ \text{kJ mol}^{-1}$

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Option 3:

$166\ \text{kJ mol}^{-1}$

Option 4:

$-83\ \text{kJ mol}^{-1}$

Correct Answer:

$41.5\ \text{kJ mol}^{-1}$

Solution:

We know the Arrhenius equation

$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A + \ln e^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{R}\left(\frac{1}{T}\right)$$

The slope of $\left(\ln k \text{ v/s } \frac{1}{T}\right)$ of the above equation -

$$m = -\frac{E_a}{R}$$

Given, m = $-5 \times 10^3\ \text{K}$ and $R = 8.314 \text{JK}^{-1}\ \text{mol}^{-1}$

$$-5 \times 10^3 = -\frac{E_a}{8.314}$$

$$E_a = 5 \times 10^3 \times 8.314\ \text{J/mol}$$

$$E_a = 41.57 \times 10^3\ \text{J/mol}$$

$$E_a \simeq 41.5\ \text{kJ/mol}$$

Hence, the answer is the option (1).

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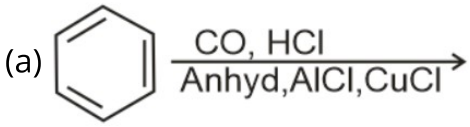
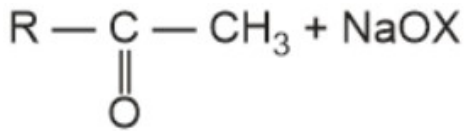
Q. 42 Match List - I with List - II

List - I	List - II
(a) 	(i) Hell - Volhard- Zelinsky reaction
(b) 	(ii) Gattermann- Koch reaction
(c) $R-CH_2-OH+R'COOH \xrightarrow{Conc.H_2SO_4}$	(iii) Haloform reaction
(d) $R-CH_2COOH \xrightarrow[(ii)H_2O]{(i)X_2/Red\ P)}$	(iv) Esterification

Choose the correct answer from the options given below.

Option 1:
(a)-(iv), (b)-(i), (c)- (ii), (d)-(iii)

Option 2:
(a) - (iii), (b)-(ii), (c)- (i), (d)-(iv)

Option 3:
(a) - (i), (b)-(iv), (c)- (iii), (d)-(ii)

Option 4:
(a) - (ii), (b)-(iii), (c)- (iv), (d)-(i)

Correct Answer:
(a) - (ii), (b)-(iii), (c)- (iv), (d)-(i)

Solution:

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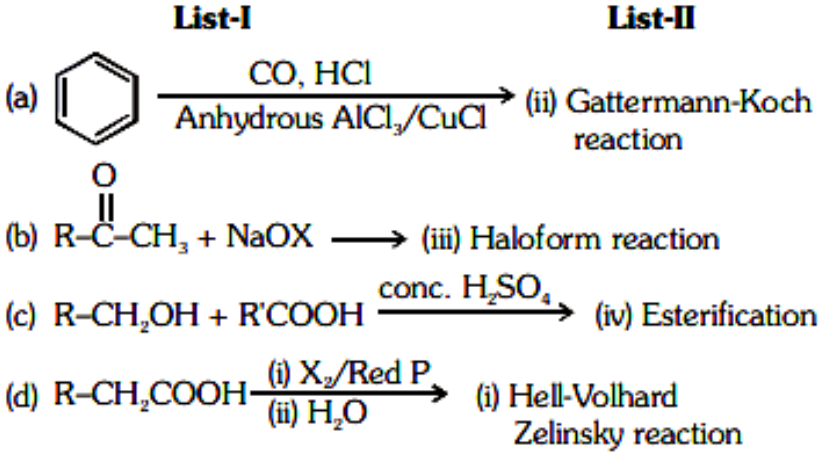

Correct answer is **option (4).**

Q. 43 For the irreversible expansion of an ideal gas under isothermal conditions, the correct option is :

Option 1:
$\Delta U = 0, \Delta S_{total} = 0$

Option 2:
$\Delta U \neq 0, \Delta S_{total} \neq 0$

Option 3:
$\Delta U = 0, \Delta S_{total} \neq 0$

Option 4:
$\Delta U \neq 0, \Delta S_{total} = 0$

Correct Answer:
$\Delta U = 0, \Delta S_{total} \neq 0$

Solution:
For irreversible expansion of an ideal gas under isothermal conditions -

We know, for a spontaneous process, $\Delta S_{\text{total}} > 0$ and since the irreversible process is always spontaneous, therefore $\Delta S_{\text{total}} > 0$.

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And $\Delta T = 0$ for the isothermal process.

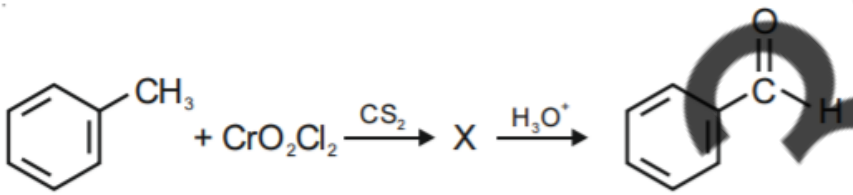
Now, $\Delta U = nC_V\Delta T$

So, $\Delta U = 0$

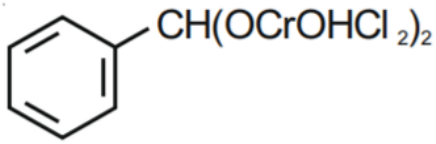
Correct option $\Delta U = 0, \Delta S_{total} \neq 0$.

Hence, the answer is the option (3).

Q. 44 The Intermediate compound 'X' in the following chemical reaction is :

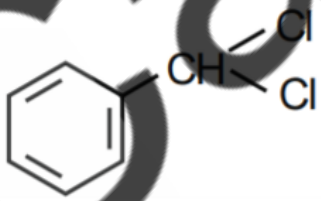


Option 1:



Option 2:

Option 3:



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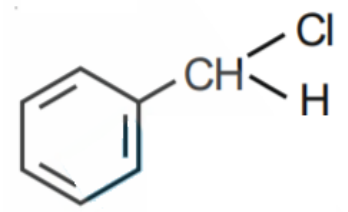
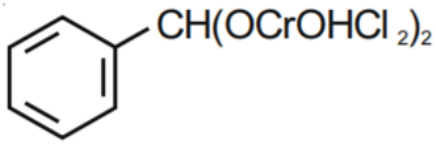
Option 4:

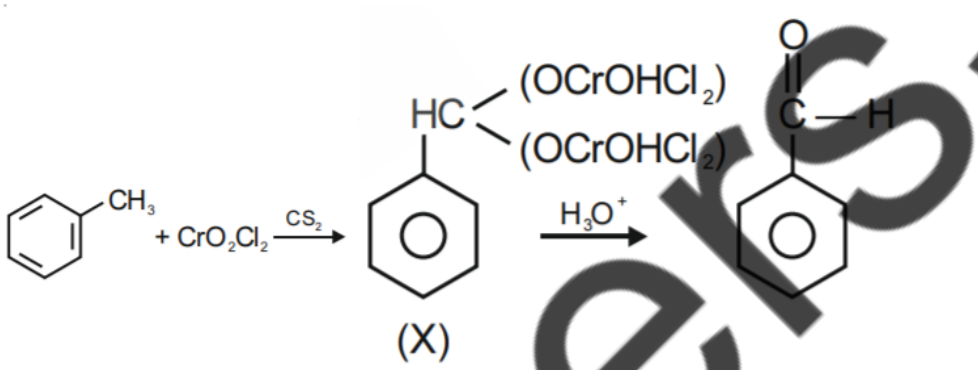


Correct Answer:



Solution:

The reaction will be -



Therefore, the correct option is (1).

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Q. 45 Match List - I with List - II.

List - I	List - II
(a) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	(i) Acid rain
(b) $HOCl(g) \overset{h\nu}{\rightarrow} \dot{O}H + \dot{C}l$	(ii) Smog
(c) $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$	(iii) Ozone depletion
(d) $NO_2(g) \overset{h\nu}{\rightarrow} NO(g) + O(g)$	(iv) Troposphere pollution

Choose the correct answer from the optionsgiven below.

Option 1:
(a) - (i), (b)-(ii), (c)-(iii),(d)-(iv)

Option 2:
(a) - (ii), (b)-(iii), (c)-(iv),(d)-(i)

Option 3:
(a) - (iv), (b)-(iii), (c)-(i),(d)-(ii)

Option 4:
(a) - (iii), (b)-(ii), (c)-(iv),(d)-(i)

Correct Answer:
(a) - (iv), (b)-(iii), (c)-(i),(d)-(ii)

Solution:
We know about these reaction -

(a) $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ - (iv) Troposphere pollution

(b) $HOCl(g) \overset{h\nu}{\rightarrow} \dot{O}H + \dot{C}l$ - (iii) Ozone depletion

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(c) $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + H_2O + CO_2$ - (i) Acid rain

(d) $NO_2(g) \overset{h\nu}{\rightarrow} NO(g) + O(g)$ - (ii) Smog

So, the correct match will be

(a) - (i), (b)-(ii), (c)-(iii),(d)-(iv)

Therefore, the correct option is (3)

Q. 46 The molar conductivity of 0.007 M acetic acid is $20\ \text{S cm}^2\text{mol}^{-1}$. What is the dissociation constant of acetic acid? Choose the correct option.

$$\left[\begin{array}{l} \Lambda^{\circ}_{H^+} = 350\ \text{S cm}^2\ \text{mol}^{-1} \\ \Lambda^{\circ}_{CH_3COO^-} = 50\ \text{S cm}^2\ \text{mol}^{-1} \end{array}\right]$$

Option 1:
$1.75 \times 10^{-4}\ \text{mol L}^{-1}$

Option 2:
$2.50 \times 10^{-4}\ \text{mol L}^{-1}$

Option 3:
$1.75 \times 10^{-5}\ \text{mol L}^{-1}$

Option 4:
$2.50 \times 10^{-5}\ \text{mol L}^{-1}$

Correct Answer:
$1.75 \times 10^{-5}\ \text{mol L}^{-1}$

Solution:
Given,

$$\Lambda_m = 20\ \text{S cm}^2\ \text{mol}^{-1}$$

$$\Lambda^{\circ}_m CH_3COOH = \Lambda^{0}_{CH_3COO^-} + \Lambda^{\circ}_{m^+}$$

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$$\Lambda_m^\circ CH_3COOH = 50 + 350 = 400\ S\ cm^2\ mol^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{20}{400} = \frac{1}{20}$$

$$K_a = \frac{C\alpha^2}{1-\alpha} \simeq C\alpha^2 = 7 \times 10^{-3} \times \left(\frac{1}{20}\right)^2$$

$$K_a = 7 \times 10^{-3} \times \frac{1}{4} \times 10^{-2}$$

$$K_a = 1.75 \times 10^{-5}\ mol\ L^{-1}$$

Hence, the answer is the option (3).

Q. 47 From the following pairs of ions which one is not an iso-electronic pair ?

Option 1:
O^{2-}, F^-

Option 2:
Na^+, Mg^{2+}

Option 3:
Mn^{2+}, Fe^{3+}

Option 4:
Fe^{2+}, Mn^{2+}

Correct Answer:
Fe^{2+}, Mn^{2+}

Solution:
The number of electrons present in the given ions-

$O^{2-} \Rightarrow 18$

$F^- \Rightarrow 18$

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$Na^+ \Rightarrow 10$

$Mg^{2+} \Rightarrow 10$

$Mn^{2+} \Rightarrow 23$

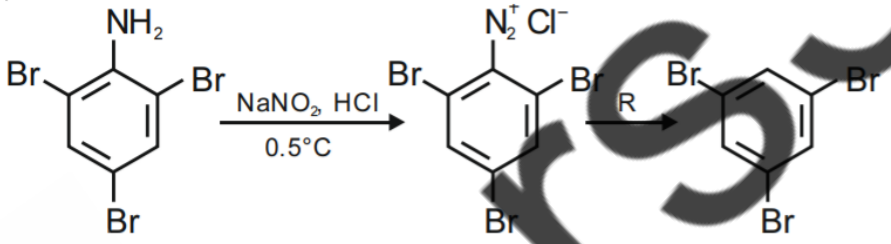
$Fe^{3+} \Rightarrow 23$

$Fe^{2+} \Rightarrow 24$

So, the pair of Fe^{2+}, Mn^{2+} ions is not iso-electronic.

Hence, the answer is the option (4).

Q. 48 The reagent 'R' in the given sequence of chemical reaction is:



Option 1:
H_2O

Option 2:
CH_3CH_2OH

Option 3:
HI

Option 4:
$CuCN/KCN$

Correct Answer:
CH_3CH_2OH

Solution:

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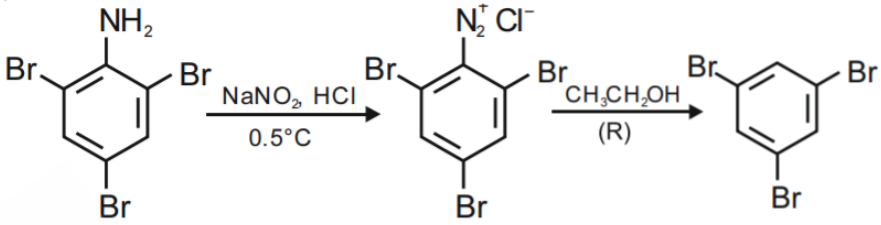

Here CH_3CH_2OH will be R.

Therefore, the correct option is (2).

Q. 49 The correct option for the value of vapour pressure of a solution at 45^oC with benzene to octane in molar ratio 3:2 is :

[At 45^oC vapour pressure of benzene is 280 mm Hg and that of octane is 420 mm Hg. Assume Ideal gas]

Option 1:
160 mm of Hg

Option 2:
168 mm of Hg

Option 3:
336 mm of Hg

Option 4:
350 mm of Hg

Correct Answer:
336 mm of Hg

Solution:
We know this formula for vapour pressure of a solution -

$$P_{Total} = \chi_A P_A + \chi_B P_B$$

At 45°C,

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vapour pressure of benzene (PA)= 280 mm Hg

vapour pressure of octane (PB) = 420 mm Hg

Given, benzene to octane in molar ratio 3:2

So,

$$\mathrm{P} = \frac{3}{3+2} \times 280 + \frac{2}{3+2} \times 420$$

$$\mathrm{P} = 168 + 168$$

$$\mathrm{P} = 336$$

Therefore, the correct option is (3).

Q. 50 $CH_3CH_2COO^-Na^+ \xrightarrow[Heat]{NaOH,+?} CH_3CH_3 + Na_2CO_3$

Consider the above reaction and identify the missing reagent/chemical.

Option 1:
B_2H_6

Option 2:
Red Phosphorus

Option 3:
CaO

Option 4:
DIBAL-H

Correct Answer:
CaO

Solution:
The reaction will be -

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$$CH_3CH_2COO^-Na^+ \xrightarrow[\text{Heat}]{NaOH+CaO} CH_3CH_3 + Na_2CO_3$$

by heating, the sodium salt of the carboxylic acid with soda lime (NaOH and CaO in the ratio of 3: 1), Alkane is produced.

Hence, the answer is the option (3).

Biology

Q. 1 Which of the following algae contains **mannitol** as reserve food material?

Option 1:
Ectocarpus

Option 2:
Graciliaria

Option 3:
Volvox

Option 4:
Ulothrix

Correct Answer:
Ectocarpus

Solution:
Ectocarpus, a type of brown algae, contains mannitol as a reserve food material. Mannitol is a sugar alcohol that serves as a storage carbohydrate in certain algae and plants.
Hence, the correct answer is option 1:

Q. 2 Amenasalism can be represented as:

Option 1:
Species A (-) : Species B (0)

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Option 2:
Species A (+) : Species B (+)

Option 3:
Species A (-) : Species B (-)

Option 4:
Species A (+) : Species B (0)

Correct Answer:
Species A (-) : Species B (0)

Solution:
Ammensalism: It is an interaction between two organisms of different species in which one species inhibits the growth of other species by secreting certain chemicals.

A(-) is inhibited

B(0) is unaffected.

Q. 3 Match List -I with List - II.

List -I	List - II
(a) Cristae	(i) Primary construction in chromosome.
(b) Thylakoids	(ii) Disc - shaped sacs in Golgi appartus
(c) Centromere	(iii) Infoldings in mitochondria
(d) Cisternae	(iv) Flattered membranes sacs in stromn in plastids

Choose the correct answer from the options given below.

Option 1:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Option 2:
(a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

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Option 3:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Option 4:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Correct Answer:
(a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

Solution:
Cristae: Infoldings in mitochondria. **To increase the capacity of the mitochondrion to synthesize ATP**, the inner membrane of mitochondria is folded to form cristae. These folds allow a much greater amount of electron transport chain enzymes and ATP synthase to be packed into the mitochondrion.

Thylakoids: are **membrane-bound compartments inside chloroplasts and cyanobacteria**. They are the site of the light-dependent reactions of photosynthesis. Thylakoids consist of a thylakoid membrane surrounding a thylakoid lumen.

Centromere: Primary construction in **chromosome**.The two chromatids of chromosomes are attached to each other by a narrow area called centromere, also called **primary constriction**. This decides the ratio of arm length called **centromeric index.**

Cisternae: Disc - shaped sacs in **Golgi appartus**. These appear in the form of fla, curved, smooth membrane structures with swollen ends. They have a definite polarity. Their **convex pole** or **proximal pole** or **cis face** is associated with nuclear envelope. The **concave pole** or **distal pole** or **trans face**, is the oldest, which give rise to the **secondary vesicles.**

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Q. 4 The plant hormone used to destroy weeds in a field is :

Option 1:
IAA

Option 2:
NAA

Option 3:
2,4 -D

Option 4:
IBA

Correct Answer:
2,4 -D

Solution:
The 2,4-dichlorophenoxyacetic acid **(2,4-D)** is **a synthetic auxin that has been historically used as one of the most common herbicides to control broadleaf weeds**. At the same time, it also has good potential to enhance storage life and quality of fruits.

Q. 5 Diadelphous stamens are found in:

Option 1:
China Rose

Option 2:
Citrus

Option 3:
Pea

Option 4:
Chia rose and citrus

Correct Answer:
Pea

Solution:
Monadelphous stamens are seen in china rose, these stamens have a single bunch of filaments.

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The plants in which the stamen or anthers are present in three or more than three bundles are known as **polyadelphous stamen**. This type of arrangement is commonly observed in **citrus fruits like lemon.**

Diadelphous stamens are found in pea, having united filaments so that they are arranged in two groups.

Q. 6 A typical angiosperm embryo sac at maturity:

Option 1:
8 - nucleate and 7-celled

Option 2:
7 - nucleate and 8-celled

Option 3:
7- nucleate and 7-celled

Option 4:
8 - nucleate and 8-celled

Correct Answer:
8 - nucleate and 7-celled

Solution:
Nucleus of functional megaspore divides by three mitotic divisions to form 8 nuclei, out of which

(i) Three of them get organised as cells at micropylar end forming egg appartus. One is **egg cell** (n) and two are **synergids (n)** or co-operative cells.

(ii) Three cells at chalazal end form **antipodals** (n) or vegetative cells of female gametophyte.

(iii) Two nuclei (one from each pole) in the centre are called **polar nuclei** (n).

As the mebryo sac matures, these polar nuclei get fused to form a **secondary nucleus** (2n) or definite nucleus.

This constitutes an embryo sac as **7-celled and 8-nucleated** structure.

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Q. 7 Which of the following is correct sequence of steps in PCR (Polymearse Chain Reaction)?

Option 1:
Denaturation, Annealing , Extenstion

Option 2:
Denaturation, Extension, Annealing

Option 3:
Extension, Denaturation, Annealing

Option 4:
Annealing, Denaturation, Extension

Correct Answer:
Denaturation, Annealing , Extenstion

Solution:
PCR is based on three simple steps required for any DNA synthesis reaction:

(1) denaturation of the template into single strands;

(2) annealing of primers to each original strand for new strand synthesis; and

(3) extension of the new DNA strands from the primers.

Q. 8 The site of perception of light in plants during photoperiodism is:

Option 1:
Shoot apex

Option 2:
Stem

Option 3:
Axillary bud

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Option 4:
Leaf

Correct Answer:
Leaf

Solution:
The site of perception of photoperiod is **leaves**. The hormone **florigen**, which is responsible for flowering, induces flowering as it migrates from leaves to shoot apices on induction of required photoperiods.

Q. 9 Which of the following is incorrect statement?

Option 1:
Mature sieve tube elements posses a conspicous nucleus and usual cytoplasmic organelles.

Option 2:
Microbodies are present both in plants and animals cells.

Option 3:
The perinuclear space forms a barrier between the materials present inside the nucleus and the cytoplasm.

Option 4:
Nuclear pores act as passage for proteins and RNA molecules in both the directions between nucleus and cytoplasm.

Correct Answer:
Mature sieve tube elements posses a conspicous nucleus and usual cytoplasmic organelles.

Solution:
Mature sieve tube elements do not have nucleus but have cytoplasm known as a**nucleated** living cells.

Q. 10 The first stable product of CO_2 fixation in sorghum is:

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Option 1:
Pyruvic acid

Option 2:
Oxaloacteic acid

Option 3:
Succinic acid

Option 4:
Phosphoglyceric acid

Correct Answer:
Oxaloacteic acid

Solution:
The first stable product of CO_2 fixation in sorghum is Oxaloacetic acid (OAA), as sorghum is a C_4 plant.

Q. 11 When gene targetting involves gene amplification is attempted in an individual's tissue to treat disease, it is known as:

Option 1:
Biopiracy

Option 2:
Gene therapy

Option 3:
Molecular diagnosis

Option 4:
Safety testing

Correct Answer:
Gene therapy

Solution:

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In gene therapy technique the absent or faulty gene is replaced by a working gene, so that the body can make the correct enzyme or protein and consequently eliminate the root cause of the disease. The first clinical gene therapy was given in 1990 to a 4- year old girl with **adenosine deaminase (ADA) deficiency.**

Q. 12 Match List -I with List - II

List - I	List -II
(a) Protoplast fusion	(i) Totipotency
(b) Plant tissue culture	(ii) Pomato
(c) Meristem culture	(iii) Somaclones
(d) Micropropagation	(iv) Virus free plants

Choose the correct answer from the options given below:

Option 1:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Option 2:
(a) - (ii), (b) - (i) , (c) - (i), (d) - (iii)

Option 3:
(a) - (iii), (b) - (iv) , (c) - (i), (d) - (ii)

Option 4:
(a) - (iv), (b) - (iii) , (c) - (ii), (d) - (i)

Correct Answer:
(a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)

Solution:
Somatic fusion, also called **protoplast fusion,** is a type of genetic modification in plants by which two distinct species of plants are fused together to form a new hybrid plant with the characteristics of both, a **somatic hybrid.**

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Plant tissue culture is defined as culturing plant seeds, organs, explants, tissues, cells, or protoplasts on a chemically defined **synthetic nutrient media** under sterile and controlled conditions of light, temperature, and humidity.

Meristem culture is defined as the tissue culture technique, which uses apical meristem with 1-3 leaf primordia to prepare clones of a plant by the vegetative propagation. This technique primarily involves the isolation of meristem by applying a V-Shape cut in the stem.

Micropropagation is **the clonal propagation of plants in closed vessels under aseptic conditions**. Inside the vessels, the plants are grown on culture media that contain nutrients and growth regulators, and are described as in vitro, which means 'in glass'. In contrast, soil-grown plants are described as in vivo.

Totipotency is defined in Wikipedia as the ability of a single cell to divide and produce all the differentiated cells in an organism, including extraembryonic tissues. Totipotent cells formed during sexual and asexual reproduction include spores and zygotes.

The **pomato** is a grafted plant that is produced by grafting together a tomato plant and a potato plant, both of which are members of the **Solanum genus** in the **Solanaceae family**.

Somaclones are the plants that are produced through tissue culture and are genetically identical to the parent plant from which they are produced and cultured.

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Q. 13 Which of the following stages of meiosis involves division of centromere?

Option 1:
Metaphase I

Option 2:
Metaphase II

Option 3:
Anaphase II

Option 4:
Telophase II

Correct Answer:
Anaphase II

Solution:
Anaphase II begins with simultaneous splitting of the centromere of each chroosome that allows them to move towards opposite poles of the cell.

Q. 14 Which of the following are not secondary metabolites in plants?

Option 1:
Morphine, codeine

Option 2:
Amino acids, glucose

Option 3:
Vinblastin, curcumin

Option 4:
Rubber, gums

Correct Answer:
Amino acids, glucose

Solution:
Amino acids and glucose are **primary metabolites** in plants.

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Primary metabolites are compounds that are directly involved in the growth and development of a plant whereas secondary metabolites are compounds produced in other metabolic pathways that, although important, are not essential to the functioning of the plant.

Q. 15 Which of the following plants is monoecious?

Option 1:
Carica papaya

Option 2:
Chara

Option 3:
Marchantia polymorpha

Option 4:
Cycas circinalis

Correct Answer:
Chara

Solution:
Monoecious or homothallic condition in Chara (green algae) is used to denote upper oogonium and lower antheridium on the same plant.

Q. 16 Mutations in plant cells can be induced by:

Option 1:
Kinetin

Option 2:
Infrared rays

Option 3:
Gamma rays

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Option 4:
Zeatin

Correct Answer:
Gamma rays

Solution:
Gamma rays are ionizing radiation and used in inducing mutations in seeds and other planting materials such as cuttings, pollen, or tissue-cultured calli.

Q. 17 During purification process for recombinant DNA technology, addition of chilled ethanol precipitates out:

Option 1:
RNA

Option 2:
DNA

Option 3:
Histones

Option 4:
Polysaccharides

Correct Answer:
DNA

Solution:
The purified DNA, after treatment with various enzymes, precipitates out after addition of chilled ethanol. This is viewed as a collection of fine threads in the suspension, and is easily collected. The process is known as **DNA spooling**.

Q. 18 Plants follow different pathways in response to environment or phases of life to form different kinds of structures.This ability is called:

Option 1:
Elasticity

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Option 2:
Flexibility

Option 3:
Plasticity

Option 4:
Maturity

Correct Answer:
Plasticity

Solution:
Plant plasticity refers to a plant's ability to adapt to and cope with changes in its environment. In contrast to animals, which are able to actively move away to avoid challenges such as a predators or a changing climate, plants have acquired the ability to biosynthesize an unprecedented array of structurally complex bioactive natural compounds with specialized roles in order to cope with environmental challenges.

For example, **genetically identical water flea (Daphnia) clones can differ in their morphology depending on whether reared in the absence or presence of a potential predator**.

Q. 19 Which of the following is not an application of PCR (Polymearse Chain Reaction)?

Option 1:
Molecular diagnosis

Option 2:
Gene amplification

Option 3:
Purification of isolated protein

Option 4:
Detection of gene mutation

Correct Answer:
Purification of isolated protein

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Solution:
Some of the **areas of application of PCR** are mentioned here:

(i) Diagnosis of pathogens.

(ii) Diagnosis of specific mutation.

(iii) DNA fingerprinting.

(iv) Detection of specific microorganisms.

(v) In prenatal diagnosis.

(vi) Diagnosis of plant pathogens.

(vii) In palaeontology.

Q. 20 The amount of nutrients, such as carbon, nitrogen, phosphorus and calcium present in the soil at any given time, is reffered as:

Option 1:
Climax

Option 2:
Climax community

Option 3:
Standing state

Option 4:
Standing crop

Correct Answer:
Standing state

Solution:
Standing state is **amount of inorganic nutrients found in an ecosystem**. It represents part of non-living matter. It circulates between living and non-living components of the ecosystem. It is being regularly depleted and replenished by the living matters. Standing crop is amount of biomass present in an ecosystem.

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Q. 21 The term used for transfer of pollen grains from anther of one plant to the stigma of a different plant which, brings genetically differnt types of pollen grains to stigma, is:

Option 1:
Xenogamy

Option 2:
Geitonogamy

Option 3:
Chasmogamy

Option 4:

Correct Answer:
Xenogamy

Solution:
Transfer of pollen grains from the anther to the stigma of same flower is known as **autogamy** or self pollination.

Geitonogamy is transfer of pollens from anther of one flower to stigma of another flower of the same plant

Xenogamy is the transfer of pollen grains from the anther to the stigma of a different plant.

Q. 22 The production of gametes by the parents, formation of zygotes, the F_1 and F_2 plants, can be understood from a diagram is called:

Option 1:
Bullet square

Option 2:
Punch square

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Option 3:
Punnett square

Option 4:
Net square

Correct Answer:
Punnett square

Solution:
The Punnett square is a square diagram that is used to predict the genotypes of a particular cross or breeding experiment. It is named after **Reginald C. Punnett**, who devised the approach in 1905. The diagram is used by biologists to determine the probability of an offspring having a particular genotype.

Q. 23 In the equation GPP - R = NPP

R represents:

Option 1:
Radiant energy

Option 2:
Retardation factor

Option 3:
Environment factor

Option 4:
Respiration losses

Correct Answer:
Respiration losses

Solution:
GPP is gross primary productivity that is the rate of total production of biomass by producers.

R is the respiratiory loss. A part of biomass produced by the plant utilized by them for their respiration.

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NPP is the balance of biomass or energy remaining after meeting the cost of respiration of producers.

Q. 24 DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as:

Option 1:
Yellow bands

Option 2:
Bright orange bands

Option 3:
Dark red bands

Option 4:
Bright blue bands

Correct Answer:
Bright orange bands

Solution:
DNA strands on a gel stained with ethidium bromide when viewed under UV radiation, appear as **Bright orange bands** as EtBr absorbs UV radiations at **210-285nm** wavelength and get excited at **605 nm** due to which it flourescence orange colour.

Q. 25 The factor that leads to founder effect in a population is:

Option 1:
Natural selection

Option 2:
Genetic recombination

Option 3:
Mutation

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Option 4:
Genetic drift

Correct Answer:
Genetic drift

Solution:
Founder effect and bottleneck effect are two kinds of Genetic drift.

Q. 26 When the centromere is situated in the middle of two equal arms of chromosomes, the chromosomes is reffered as:

Option 1:
Metacentric

Option 2:
Telocentric

Option 3:
Sub - metacentric

Option 4:
Acrocentric

Correct Answer:
Metacentric

Solution:
On the basis of the location of the centromere, chromosomes are classified into four types: **metacentric, submetacentric, acrocentric, and telocentric**.

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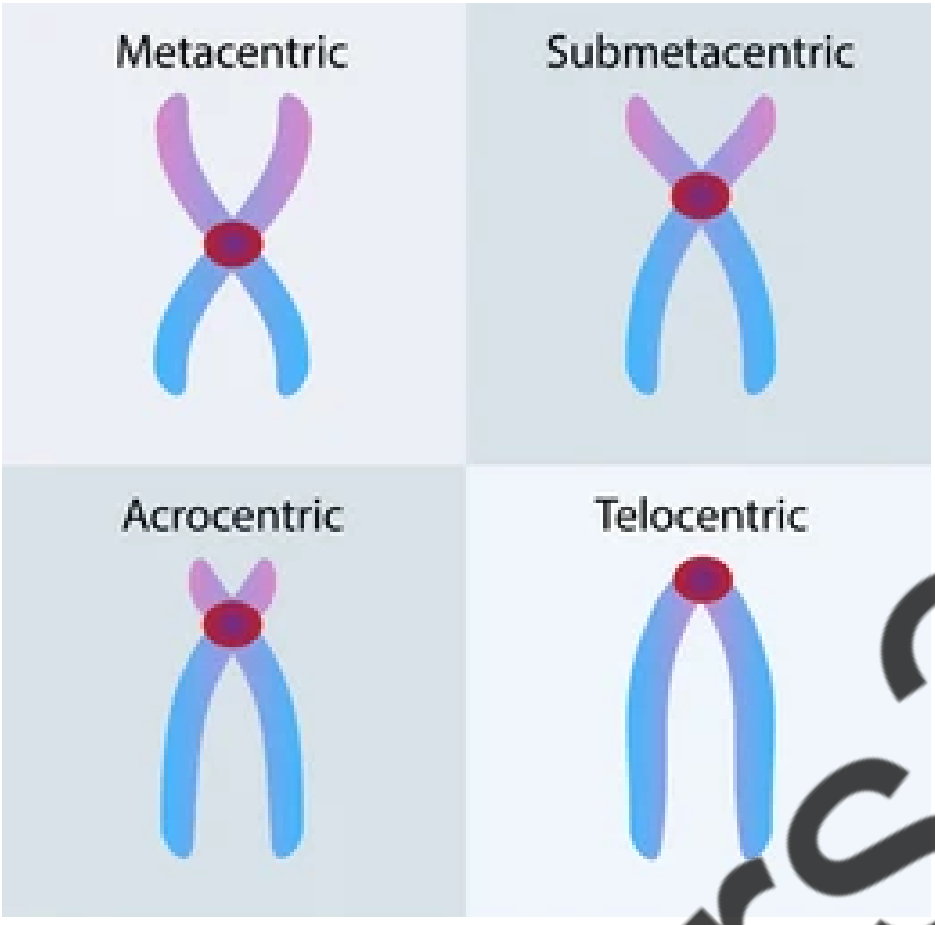

Q. 27 Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as:

Option 1:
Homosorus

Option 2:
Heterosorus

Option 3:
Homosporous

Option 4:
Heterosporous

Correct Answer:
Heterosporous

Solution:

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Heterospory is the production of spores of two different sizes and sexes by the sporophytes of land plants**.** The smaller of these, the microspore, is male and the larger megaspore is female.

Q. 28 Match List - I with List - II

List - I	List - II
(a) Cohesion	(i) More attraction in liquid plants
(b) Adhesion	(ii) Mutual attraction among water molecules
(c) Surface tension	(iii) Water loss in liquid phase
(d) Guttation	(iv) Attraction towards polar surfaces

Choose the correct answer from the options given below:

Option 1:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 2:
(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

Option 3:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Option 4:
(a) - (ii), (b) - (i), (c) - (iv), (d) - (iii)

Correct Answer:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Solution:
Cohesion, adhesion and surface tension are important proprties of water as they help the water from a tree's roots to be transported all the way to its leaves.

Cohesion holds hydrogen bonds together to create **surface tension** on water. Since water is attracted to other molecules, **adhesive forces** pull the water toward other molecules.

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Q. 29 Gemmae are present in:

Option 1:
Liverworts

Option 2:
Brown algae

Option 3:
Red algae

Option 4:
Blue - green algae

Correct Answer:
Liverworts

Solution:
Most **liverworts** can reproduce asexually by means of **gemmae**, which are disks of tissues produced by the gametophytic generation. The gemmae are held in special organs known as **gemma cups** and are dispersed by rainfall.

Q. 30 Which of the follwoing algae produces Carrageenan ?

Option 1:
Green algae

Option 2:
Brown algae

Option 3:
Red algae

Option 4:
Blue - green algae

Correct Answer:
Red algae

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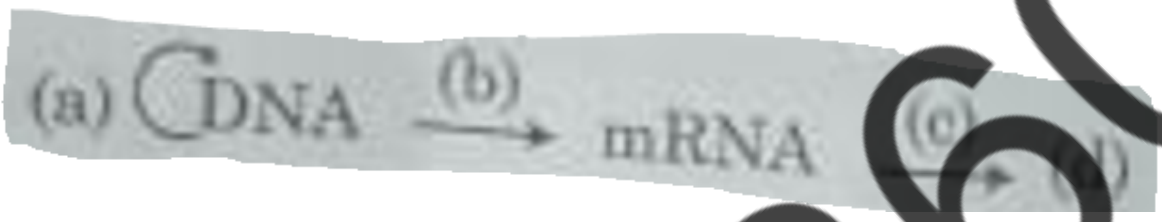
Solution:
Carrageenan is a **phycocolloids** produce in **red algae (Rhodophyceae)**. It is used as a clearing agent in liquors, leather finishing and as a emulsifier in chocolates, ice-creams, toothpastes, paints etc.

Q. 31 Complete the flow chart on central dogma.



Option 1:
(a) - Replication ; (b) - Transcription; (c) - Transduction; (d) - Protein

Option 2:
(a) -Translation; (b) - Replication ; (c) - Transcription ; (d) - Transduction

Option 3:
(a) -Replication; (b) - Transcription ; (c) - Translation ; (d) - Protein

Option 4:
(a) -Transduction ; (b) - Translation ; (c) - Replication ; (d) - Protein

Correct Answer:
(a) -Replication; (b) - Transcription ; (c) - Translation ; (d) - Protein

Solution:
The central dogma of molecular biology is **an explanation of the flow of genetic information within a biological system**. It is often stated as "DNA makes RNA, and RNA makes protein".

It was first stated by **Francis Crick** in 1957, then published in 1958.

Q. 32 Which of the following statements is not correct?

Option 1:
Pyramid of biomass in sea is generally inverted.

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Option 2:
Pyramid of biomass in sea is generally upright.

Option 3:
Pyramid of energy is always upright.

Option 4:
Pyramid of numbers in a grassland ecosystem is upright.

Correct Answer:
Pyramid of biomass in sea is generally upright.

Solution:
The pyramid of biomass in the sea is inverted because **the amount of biomass is least at the base of the pyramid and the amount of biomass is maximum at the apex of the pyramid**.

Eg: Biomass of zooplanktons is higher than that of phytoplanktons as life span of former is longer and the latter multiply much faster though having shorter life span. A number of generations of phytoplanktons may thus be consumed by single generation of zooplanktons.

Q. 33 Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival?

Option 1:
Resource partitioning

Option 2:
Competitive release

Option 3:
Mutualism

Option 4:
Predation

Correct Answer:
Resource partitioning

Solution:

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Resource partitioning or niche differentiation (**a process of natural selection that will force competitors to use resources differently**) is a way to avoid competition between species and allow for coexistence.

Species might evolve mechanism to live in the same niche by changing the **feeding time** or **foraging patterns.**

Q. 34 Match List -I with List -II

List - I	List - II
(a) Lenticels	(i) Phellogen
(b) Cork cambium	(ii) Suberin deposition
(c) Secondary cortex	(iii) Exchange of gases
(d) Cork	(iv) Phelloderm

Choose the correct answer from the options given below:

Option 1:
(a) - (iv), (b) - (i), (c) - (iii), (d) - (ii)

Option 2:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Option 3:
(a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)

Option 4:
(a) - (iv), (b) - (ii), (c) - (i), (d) - (iii)

Correct Answer:
(a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

Solution:
Lenticels: In cork layer the lenticels are present which are meant for gaseous exchange. In cork, lenticels have loosely aggranged cells called **complementary cells** with intercelluar spaces.

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Cork cambium is **a** tissue found in many vascular plants as a part of the epidermis. It is one of the many layers of bark, between the cork and primary phloem. The cork cambium is a lateral meristem and is responsible for secondary growth that replaces the epidermis in roots and stems.

Secondary cortex is also known as **phelloderm**, produces from cork cambium on innerside of stem. Cells of phelloderm are thin walled, living and store food.

Cork also known as **phellem** produces from cork cambium on the outerside of the stem. The cells of phellem are dead, suberized and impervious to water.

Cork (phellem) + Cork cambium (phellogen) + Secondary cortex (Phelloderm)= Periderm.

Q. 35 Match the List - I and List -II

List - I	List - II
(a) Cells with active cell division capacity	(i) Vascular tissue
(b) Tissue having all cells similar in structure and function	(ii) Meristematic tissue
(c) Tissue having different types of cells	(iii) Sclereids
(d) Dead cells with highly thickned walls and narrow lumen	(iv) Simple tissue

Select the correct answer from the options given below:

Option 1:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Option 2:
(a) - (iv), (b) - (iii), (c) - (iii), (d) - (i)

Option 3:
(a) - (i), (b) - (ii), (c) - (iii), (d) - (iv)

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Option 4:
(a) - (iii), (b) - (ii), (c) - (iv), (d) - (i)

Correct Answer:
(a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

Solution:
Cells with active cell division capacity- **Meristematic tissue**

Tissue having all cells similar in structure and function- **Simple tissue**

Tissue having different types of cells- **Vascular tissue**

Dead cells with highly thickned walls and narrow lumen- **Sclereids**

Q. 36 Match Column - I with Column - II

Column - I	Column - II
(a) Nitrococcus	(i) Denitrification
(b) Rhizobium	(ii) Conversion of ammonia to nitrite
(c) Thiobacillus	(iii) Conversion of nitrite to nitrate
(d) Nitrobacter	(iv) Convesion of atmospheric nitrogen to ammonia

Chhose the correct answer from options given below:

Option 1:
(a) - (ii), (b) - (iv). (c) - (i), (d) - (iii)

Option 2:
(a) - (i), (b) - (ii). (c) - (iii), (d) - (iv)

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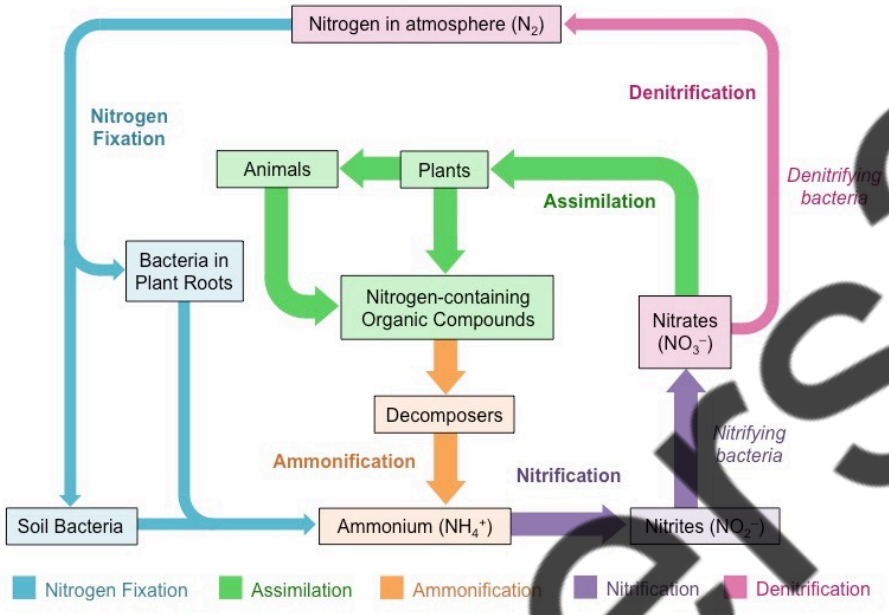
Option 3:
(a) - (iii), (b) - (i). (c) - (iv), (d) - (ii)

Option 4:
(a) - (iv), (b) - (iii). (c) - (ii), (d) - (i)

Correct Answer:
(a) - (ii), (b) - (iv). (c) - (i), (d) - (iii)

Solution:



Q. 37 Match List - I with List - II

List - I	List - II
(a) S phase	(i) Proteins and synthesized
(b) G_2 phase	(ii) Inactive phase
(c) Quiescent stage	(iii) Interval between mitosis and initaiation of DNA replication
(d) G_1 phase	(iv) DNA replication

Choose the correct answer from the option given below:

Option 1:
(a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

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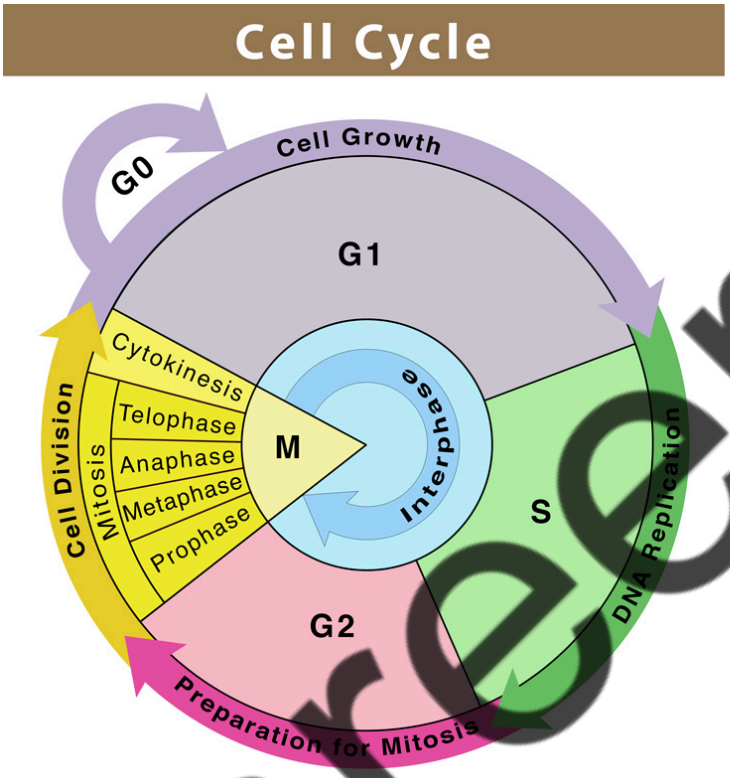
Option 2:
(a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

Option 3:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Option 4:
(a) - (ii), (b) - (iv), (c) - (iii), (d) - (i)

Correct Answer:
(a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

Solution:



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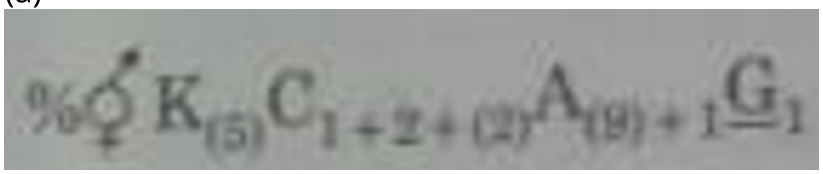
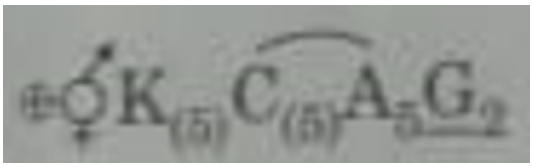
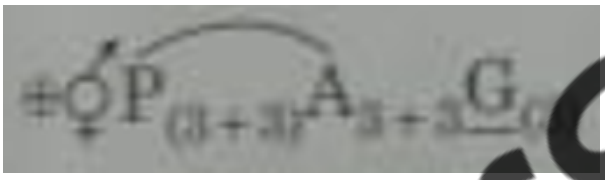
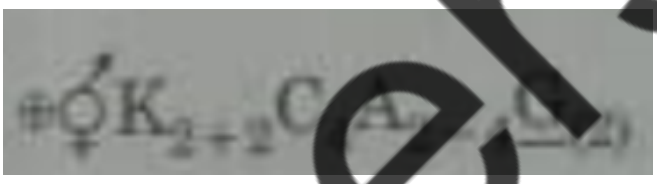
Q. 38 Match Column I with Column -II

Column - I	Column -II
(a) 	(i) Brassicaceae
(b) 	(ii) Liliaceae
(c) 	(iii) Fabaceae
(d) 	(iv) Solanaceae

Select the correct answer from the options given below:

Option 1:
(a) - (iii) , (b) - (iv), (c)- (ii), (d) - (i)

Option 2:
(a) - (i) , (b) - (ii), (c)- (iii), (d) - (iv)

Option 3:
(a) - (ii) , (b) - (iii), (c)- (iv), (d) - (i)

Option 4:
(a) - (iv) , (b) - (ii), (c)- (i), (d) - (iii)

Correct Answer:
(a) - (iii) , (b) - (iv), (c)- (ii), (d) - (i)

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Solution:
A Floral formula is **a notation for representing the structure of particular types of flowers**. Such notations use numbers, letters and various symbols to convey significant information in a compact form.

Q. 39 Which of the following statements is incorrect?

Option 1:
During aerobic respiration, role of oxygen is limited to the terminal stage.

Option 2:
In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and $FADH_2$ gives rise to 3 ATP molecules.

Option 3:
ATP is synthesized through complex V.

Option 4:
Oxidation - reduction reactions produces proton gradient in respiration.

Correct Answer:
In ETC (Electron Transport Chain), one molecule of NADH + H^+ gives rise to 2 ATP molecules, and $FADH_2$ gives rise to 3 ATP molecules.

Solution:
Oxidation of one molecule of NADH gives rise to 3 molecules of ATP while that of one molecule of $FADH_2$ produces 2 molecules of ATP.

Q. 40 In the exponential growth equation

$N_t = N_0 e^{rt}$ represents:

Option 1:
The base number logarithms

Option 2:
The base of exponential logarithms

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Option 3:
The base of natural logarithms

Option 4:
The base of general logarithms

Correct Answer:
The base of natural logarithms

Solution:
The integral equation for exponential growth is $\mathbf{N_t=N0^{ert}}$. Where N_t=Population density after time t,No =Population density at time zero,r=intrinsic rate of natural increase,e=the base of natural logarithms.

Q. 41 Which of the following statements is incorrect?

Option 1:
Both ATP and NADPH + H^+ are synthesized during non-cyclic photophosphorylation

Option 2:
Stroma lamellae have PS I only and lack NADP reductase.

Option 3:
Grana lamellae have both PS I and PS II

Option 4:
Cyclic photophosphorylation involves both PS I and PS II

Correct Answer:
Cyclic photophosphorylation involves both PS I and PS II

Solution:
When only PS I is functional, the electron is circulated within the photosystem and the phosphorylation occurs, due to the cyclic flow of electrons. The membrane and lamella of the grana have both PS I and PS II, the stroma lamella membrane lack PS II as well as NADP reductase enzyme.

Q. 42 What is the role of RNA polymerase III in the process of transcription in eukaroytes?

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Option 1:
Transcribe rRNAs (28S, 18S and 5.8S)

Option 2:
Transcribes tRNA, 5S rRNAA and snRNA.

Option 3:
Transcribes precussor of mRNA

Option 4:
Transcribes only snRNAs

Correct Answer:
Transcribes tRNA, 5S rRNAA and snRNA.

Solution:
In eukaryote cells, RNA polymerase III (also called Pol III) **transcribes DNA to synthesize ribosomal 5S rRNA, tRNA and other small RNAs**. The genes transcribed by RNA Pol III fall in the category of "housekeeping" genes whose expression is required in all cell types and most environmental conditions.

Q. 43 Now a days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hydridise its complimentary DNA in a clone of cells, followed by its detection using autoradiography because:

Option 1:
mutated gene partially appears on a photographic film

Option 2:
mutated gene completely and clearly appears on photographic films

Option 3:
mutated gene does not appear on a photographic film as the probe has no complumentary with it.

Option 4:
mutated gene does not appear on photographic film as the probe has complimentary with it.

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Correct Answer:
mutated gene does not appear on photographic film as the probe has complimentary with it.

Solution:
A single-stranded DNA or RNA, tagged with a radioactive molecule (probe) is allowed to hybridise to its complementary DNA in a clone of cells followed by detection using autoradiography. The clone having the mutated gene will hence not appear on the photographic film, because the probe will not have complementarity with the mutated gene.

Q. 44 Which of the following statements is correct?

Option 1:
Fusion of two cells is called Karyogamy

Option 2:
Fusion of protoplasm between two motile on non -motile gametes is called plasogamy

Option 3:
Organism that depend on the living plants are called saprophytes.

Option 4:
Some of the organisms can fix atmospheric nitrogen in specialized cells called sheath cells.

Correct Answer:
Fusion of two cells is called Karyogamy

Solution:
Plasmogamy, **the fusion of two protoplasts (the contents of the two cells), brings together two compatible haploid nuclei**. At this point, two nuclear types are present in the same cell, but the nuclei have not yet fused.

Some of the organisms like **Nostoc and Anabaena** can fix atmospheric nitrogen in specialized cells called **Heterocyst**.

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Q. 45 Match List - I with List - II

List - I	List - II
(a) Protein	(i) C = C double bond
(b) Unsaturated fatty acid	(ii) Phosphodiester bonds
(c) Nucleic acid	(iii) Glycosidic bonds
(d) Polysaccharide	(iv) Peptide bond

Choose the correct answer from the options given below:

Option 1:
a - iv, b - i, c - ii , d- iii

Option 2:
a -i, b -iv, c -iii , d-ii

Option 3:
a -ii, b -i, c-iv, d - iii

Option 4:
a -iv, b - iii, c - i, d - ii

Correct Answer:
a - iv, b - i, c - ii , d- iii

Solution:
Protein- peptide bond: A **covalent bond** joining the α-amino group of one amino acid to the carboxyl group of another with the loss of a water molecule.

Unsaturated fatty acid- C=C double bond

Nucleic acid- **Phosphodiester bonds**: A phosphodiester bond occurs when exactly two of the hydroxyl groups in phosphoric acid react with hydroxyl groups on other molecules to form two ester bonds.

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Polysaccharide - **Glycosidic bonds**: glycosidic bond or glycosidic linkage is a type of covalent bond that joins a carbohydrate molecule to another group, which may or may not be another carbohydrate.

Q. 46 Select the correct pair

Option 1:
Large colorless empty cells in the epidemis of grass leaves - Subsidiary leaves

Option 2:
In dicot leaves, vascular bundles are surrounded by large thick walled cells - Conjuctive tissue

Option 3:
Cells of medullary rays yhat form part of cambial ring - Interfascicular cambium

Option 4:
Loose parenchyma cells rupturing epidermis and forming a lens-shaped opening in bark - Spongy parenchyma

Correct Answer:
Cells of medullary rays yhat form part of cambial ring - Interfascicular cambium

Solution:
Large colorless with thin flexible walls, empty cells in the epidemis of some grasses like Poa, Aropyron, Maize etc., epidermal cells are known as **Bulliform or motor cells.** These cells help in rolling and unrolling of leaves.

In dicot leaves, vascular bundles are surrounded by large thick walled cells known as **bundle sheath cells.**

Loose parenchyma cells rupturing epidermis and forming a lens-shaped opening in bark are called **lenticels**, which are meant for gaseous exchange.

Q. 47 DNA fingerprinting involves identifying differences in some specific regions in DNA sequence, called as:

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Option 1:
Satellite DNA

Option 2:
Repetitive DNA

Option 3:
Single nucleotides

Option 4:
Polymorphic DNA

Correct Answer:
Repetitive DNA

Solution:
Repetitive DNA: Stretches of DNA that repeat themselves throughout a genome, either in tandem or interspersed along the genome. These stretches can comprise up to fifty percent or more of an organism's DNA.Hence, the correct option is (b).

Q. 48 Plasmid pBR322 has PstI restriction enzyme site within gene amp^R that confers ampicillin resistance.If this enzymes is used for inserting a gene β-galactoside production and the recombinant plasmid is inserted in an E.*coli* strain

Option 1:
it will not be able to confer ampicilin resistance to the host cell

Option 2:
the transformed cells will have the ability to resist ampicillin as well as produces β -galactoside

Option 3:
it will lead to lysis of host cell

Option 4:
it will be able to produce a novel protein with dual ability

Correct Answer:
it will not be able to confer ampicilin resistance to the host cell

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Solution:
Since the gene is inserted at Pst I of ampR region of pBR322, the ampR gene is inactivated which is called as **insertional inactivation.** Hence the genetically modified E. coli strain will not be able to confer ampicillin resistance.

Q. 49 Identify the correct statement:

Option 1:
In capping, methyl guanosine triphosphate is added to the 3' end of hnRNA

Option 2:
RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

Option 3:
The coding strand in a transcription unit is copied to an mRNA

Option 4:
Split gene arrangement is characteristic of prokaryotes.

Correct Answer:
RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

Solution:
In the cell, Rho binds to untranslated naked RNAs and terminates the synthesis of mRNA at the end of a significant number of operons.

Q. 50 In some members of which of the following pairs of families, pollen grains retain their viability for months after release?

Option 1:
Poaceae: Rosaceae

Option 2:
Poaceae: Leguminosae

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Option 3:
Poaceae: Solanaceae

Option 4:
Rosaceae : Leguminosae

Correct Answer:
Rosaceae : Leguminosae

Solution:
In dicot families like **Solanaceae, Rosaceae and Leguminosae** the viability of pollen grains remain months together after their release.

Q. 51 Which of the following characteristics is incorrect with respect to cockroach ?

Option 1:
A ring of gastric caeca is present at the junction of midgut and hind gut

Option 2:
Hypopharynx lies within the cavity enclosed by the mouth parts.

Option 3:
In females, 7th , 9th sterna together form a genital pouch

Option 4:
10th abdominal segment in both sexes, bears a pair of anal cerci.

Correct Answer:
A ring of gastric caeca is present at the junction of midgut and hind gut

Solution:
Correct option is (a)

-Hepatic Caecae is between foregut and midgut

- Malphigian tubules is between midgut and hindgut.

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Q. 52 Match List - I with List - II.

List - I		List - II	
(a)	Metamerism	(i)	Coelenterata
(b)	Canal system	(ii)	Ctenophora
(c)	Comb plates	(iii)	Annelida
(d)	Cnidoblasts	(iv)	Porifera

Choose the correct answer from the option given below.

Option 1:
(a) (b) (c) (d)
(iv) (iii) (i) (ii)

Option 2:
(a) (b) (c) (d)
(iii) (iv) (i) (ii)

Option 3:
(a) (b) (c) (d)
(iii) (iv) (ii) (i)

Option 4:
(a) (b) (c) (d)
(iv) (i) (ii) (iii)

Correct Answer:
(a) (b) (c) (d)
(iii) (iv) (ii) (i)

Solution:
Correct option is (c)

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Metamerism - Segmented body(Annelids)

Canal System - Porifera (Sponges)

Combi plates - Ctenophora

Cnidoblasts - Coelentetrata (Cnidoblasts)

Q. 53 A specific recognition sequence identified by endonucleases to make cuts at specific positions within the DNA is :

Option 1:
Degenerate primer sequence

Option 2:
Okazaki sequences

Option 3:
Palindromic Nucleotide sequences

Option 4:
Poly (A) tail sequences

Correct Answer:
Palindromic Nucleotide sequences

Solution:
Palindrome sequence reads same from both the directions.It is specific recognition sequence identified by endonucleases to cut the DNA strands at the specific positions.

Q. 54 The fruit fly has 8 chromosomes (2n) in each cell. During interphase of Mitosis if the number of chromosomes at G_1 phase is 8, what would be the number of chromosomes after S phase ?

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Option 1:
8

Option 2:
16

Option 3:
4

Option 4:
32

Correct Answer:
8

Solution:
Correct option is (a)

It will be same as that of G1, hence answer is 8.

Q. 55 Persons with 'AB' blood group are called as "Universal recipients". This is due to :

Option 1:
Absence of antigens A and B on the surface of RBCs

Option 2:
Absence of antigens A and B in plasma

Option 3:
Presence of antibodies, anti- A and anti- B on RBCs

Option 4:
Absence of antibodies, anti-A and anti- B, in plasma

Correct Answer:
Absence of antibodies, anti-A and anti- B, in plasma

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Solution:
Correct option is (d)

ABO blood group is on the basis of Antigens present on the RBC surface.

It is either A blood group B, O or AB

AB blood group is due to the presence of two antigens A. Due to this no antibodies are present.

Q. 56 Which of the following statements wrongly represents the nature of smooth muscle?

Option 1:
These muscle have no striations

Option 2:
They are involuntry muscles

Option 3:
Communication among the cells is performed by intercalated discs

Option 4:
These muscles are present in the wall of blood vessels

Correct Answer:
Communication among the cells is performed by intercalated discs

Solution:
Correct option is (c)

Intercalated discs are feature of cardiac muscles only.

Q. 57 Which one of the following belongs to the family Muscidae?

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Option 1:
Fire fly

Option 2:
Grasshopper

Option 3:
Crockroach

Option 4:
House fly

Correct Answer:
House fly

Solution:
Correct option is (d)

House fly

Q. 58 Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes ?

Option 1:
DNA dependnet DNA polymerase

Option 2:
DNA dpendent RNA polymerase

Option 3:
DNA Ligase

Option 4:
DNase

Correct Answer:
DNA dpendent RNA polymerase

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Solution:

- In the process of transcription in prokaryotes, DNA-dependent RNA polymerase is the enzyme responsible for catalyzing initiation, elongation, and termination. It binds to the DNA template and synthesizes an RNA molecule complementary to the DNA strand, resulting in the production of RNA transcripts. Therefore, **option 2 is the correct answer.**
- Option 1 is incorrect because DNA-dependent DNA polymerase is involved in DNA replication.
- Option 3 is incorrect DNA ligase joins DNA fragments together
- Option 4 is incorrect DNase degrades DNA.

Q. 59 Which of the following RNAs is not required for the synthesis of protein ?

Option 1:
mRNA

Option 2:
tRNA

Option 3:
rRNA

Option 4:
aiRNA

Correct Answer:
aiRNA

Solution:
Correct option is (d)

mRNA for transcription tRNA and rRNA for translation.

Q. 60 For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection?

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Option 1:
Western Blotting Technique

Option 2:
Southern Blotting Technique

Option 3:
ELISA Technique

Option 4:
Hybridization Technique

Correct Answer:
ELISA Technique

Solution:
Correct option is (b)

Southern Blotting Technique

Q. 61 Erythropoietin hormone which stimulates R.B.C.formation is produced by :

Option 1:
Alpha cells of pancreas

Option 2:
The cells of rostral adenohypophysis

Option 3:
The cells of bone marrow

Option 4:
cells of the kidney

Correct Answer:
cells of the kidney

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Solution:
Correct option is (d)

Erythioprotein helps in the process of Erythroprocess.

Q. 62 The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are :

Option 1:
$pO_2 = 104 \; and \; pCO_2 = 40$

Option 2:
$pO_2 = 40 \; and \; pCO_2 = 45$

Option 3:
$pO_2 = 95 \; and \; pCO_2 = 40$

Option 4:
$pO_2 = 160 \; and \; pCO_2 = 80$

Correct Answer:
$pO_2 = 104 \; and \; pCO_2 = 40$

Solution:
Correct option is (a)

$pO_2 = 104 \; and \; pCO_2 = 40$

Q. 63 In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ?

Option 1:
50%

Option 2:
75%

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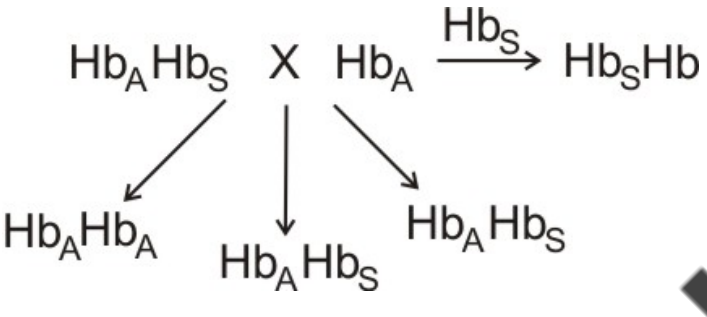
Option 3:
25%

Option 4:
100%

Correct Answer:
25%

Solution:
Correct option is (c)



Sickle cell is Autosomal receive. Only 25% of progeny is affected.

Q. 64 Which stage of meiotic prophase shows terminalisation of chiasmata as its distinctive feature?

Option 1:
Leptotene

Option 2:
Zygotene

Option 3:
Diakinesis

Option 4:
Pachytene

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Correct Answer:
Diakinesis

Solution:
Correct option is (c)

Terminalisation of Chiasmata is a feature of Diakeneses.

Q. 65 Which one of the following organisms bear hollow and pneumatic long bones ?

Option 1:
Neophron

Option 2:
Hemidactylus

Option 3:
Macropus

Option 4:
Ornithorhynchus

Correct Answer:
Neophron

Solution:
Correct option is (a)

This is a feature of Class Aves.(birds)

Neophron is Vulture

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Q. 66 Match the following :

List - I		List - II	
(a)	Physalia	(i)	Pearl oyester
(b)	Limulus	(ii)	Portuguese Man of War
(c)	Ancylostoma	(iii)	Living fossil
(d)	Pinctada	(iv)	Hookworm

Choose the correct answer from the options given below.

Option 1:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

Option 2:
(a) (b) (c) (d)
(iv) (i) (iii) (ii)

Option 3:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Option 4:
(a) (b) (c) (d)
(i) (iv) (iii) (ii)

Correct Answer:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Solution:
Correct option is (c)

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Physalia - Portuguese Man of War

Limulus - Living fossil

Ancylostoma - Hookworm

Pinctada - Pearl oyster

Q. 67 Which of the following is not an objective of Biofortification in crops ?

Option 1:
Improve protein content

Option 2:
Improve resistance to diseases

Option 3:
Improve vitamin content

Option 4:
and mineral content

Correct Answer:
Improve resistance to diseases

Solution:
Improving resistance to disease in crop is not an objective of Biofortification of crops.

Hence, the correct otpion is (b).

Q. 68 If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine, and Cytosine in it?

Option 1:
$T:20:G:30:C:20$

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Option 2:

$T : 20 : G : 20 : C : 30$

Option 3:

$T : 30 : G : 20 : C : 20$

Option 4:

$T : 20 : G : 25 : C : 25$

Correct Answer:

$T : 30 : G : 20 : C : 20$

Solution:

The correct option is (c)

$A = T$ hence $T = 30\%$

$A + T = 60\%$

$G + C = 100 - 60 = 40$

$\therefore G = C = 20\%$

Q. 69 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Vaults	(i)	Entry of sperm through the Cervix is blocked
(b)	IUDs	(ii)	Removal of Vas deferens
(c)	Vasectomy	(iii)	Phagocytosis of sperms within the Uterus
(d)	Tubectomy	(iv)	Removal of the fallopian tube

Choose the correct answer from the options given below.

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Option 1:

(a) (b) (c) (d)

(iv) (ii) (i) (iii)

Option 2:

(a) (b) (c) (d)

(i) (iii) (ii) (iv)

Option 3:

(a) (b) (c) (d)

(ii) (iv) (iii) (i)

Option 4:

(a) (b) (c) (d)

(iii) (i) (iv) (ii)

Correct Answer:

(a) (b) (c) (d)

(i) (iii) (ii) (iv)

Solution:

Correct option is (b)

Vaults - Entry of sperm through Cervix is blocked

IUDs - Phagocytosis of sperms within the Uterus

Vasectomy - Removal of Vas deferens

Tubectomy - Removal of fallopian tube

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Q. 70 Veneral diseases can spread through:

(a) Using sterile needles

(b) Transfusion of blood from infected person

(c) Infected mother of foetus

(d) Kissing

(e) Inheritance

Choose the correct answer from the options given below.

Option 1:
(a), (b) and (c) only

Option 2:
(b), (c) and (d) only

Option 3:
(b) and (c) only

Option 4:
(a) and (c) only

Correct Answer:
(b) and (c) only

Solution:
Veneral diseases are STD's. Hence, the Correct option is (c)

Q. 71 The centriole undergoes duplication during :

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Option 1:
S- phase

Option 2:
Prophase

Option 3:
Metaphase

Option 4:
G_2 Phase

Correct Answer:
S- phase

Solution:
Correct option is (a)

Centriole duplicates during S Phase

Q. 72 Receptors for sperm binding in mammals are present on :

Option 1:
Corona radiate

Option 2:
Vitelline membrane

Option 3:
Perovitelline space

Option 4:
Zona pellucida

Correct Answer:
Zona pellucida

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Solution:
Correct option is (d)

Zp (2) and ZP (3) are the receptors on Zona.

Q. 73 Dobson units are used to measure thickness of :

Option 1:
CFCs

Option 2:
Stratosphere

Option 3:
Ozone

Option 4:
Troposphere

Correct Answer:
Troposphere

Solution:
Correct option is (d)

Troposphere is a part of Ozone

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Q. 74 Match **List -I** with **List - II**.

List - I		**List - II**	
(a)	Aspergillus niger	(i)	Acetic Acid
(b)	Acetobacter aceti	(ii)	Lactic Acid
(c)	Clostridium butylicum	(iii)	Citric Acid
(d)	Lactobacillus	(iv)	Butyric Acid

Choose the correct answer from the options given below.

Option 1:
(a) (b) (c) (d)
(iii) (i) (iv) (ii)

Option 2:
(a) (b) (c) (d)
(i) (ii) (iii) (iv)

Option 3:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

Option 4:
(a) (b) (c) (d)
(iv) (ii) (i) (iii)

Correct Answer:
(a) (b) (c) (d)
(iii) (i) (iv) (ii)

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Solution:
Correct option is (a)

Aspergillus niger	(iii)	Citric Acid
Acetobacter aceti	(i)	Acetic Acid
Clostridium butylicum	(iv)	Butyric Acid
Lactobacillus	(ii)	Lactic Acid

Q. 75 Identify the **incorrect** pair.

Option 1:
Alkaloids - Codeine

Option 2:
Toxin - Abrin

Option 3:
Lectins - Concanavalin A

Option 4:
Drugs - Ricin

Correct Answer:
Drugs - Ricin

Solution:
Correct option is (d)

Ricin is s toxin

Q. 76 The organelles that are included in the endomembrane system are :

Option 1:
Endoplasmic reticulum, Mitochondria, Ribosomes and Lysosomes

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Option 2:
Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Option 3:
Golgi complex,Mitochondria,Ribosomes and Lysosomes

Option 4:
Golgi complex, Endoplasmic reticulum,Mitochondria and Lysosomes

Correct Answer:
Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Solution:
Correct option is (b)

Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles

Q. 77 Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as :

Option 1:
Arthritis

Option 2:
Muscular dystrophy

Option 3:
Myasthenia gravis

Option 4:
Gout

Correct Answer:
Myasthenia gravis

Solution:
Correct option is (c)

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Myasthenia gravis

Q. 78 With regard to insulin choose correct options.

(a) C-peptide is not present in mature insulin.

(b) The insulin produced by rDNA technology has C-peptide.

(c) The pro-insulin has C-peptide

(d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the **correct** answer from the options given below.

Option 1:
(b) and (d) only

Option 2:
(b) and (c) only

Option 3:
(a), (c) and (d) only

Option 4:
(a) and (d) only

Correct Answer:
(a), (c) and (d) only

Solution:
Correct option is (c)

(a) C-peptide is not present in mature insulin.

(c) The pro-insulin has C-peptide

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(d) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Q. 79 During the process of gene amplification using PCR, if very high temperature is not maintained in the beginning, then which of the following steps of PCR will be affected first ?

Option 1:
Annealing

Option 2:
Extension

Option 3:
Denaturation

Option 4:
Ligation

Correct Answer:
Denaturation

Solution:
Correct option is (c)

Denaturation is the first step in PCR process

Q. 80 Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins ?

Option 1:
Thrombin

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Option 2:
Renin

Option 3:
Epinephrine

Option 4:
Thrombokinase

Correct Answer:
Thrombin

Solution:
Correct option is (a)

Thrombin

Q. 81 Succus entericus is referred to as :

Option 1:
Pancreatic juice

Option 2:
Intestinal juice

Option 3:
Gastric juice

Option 4:
Chyme

Correct Answer:
Intestinal juice

Solution:
Correct option is (b)

Intestinal juice

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Q. 82 Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

Option 1:
High pO_2, low pCO_2, less H^+, lower temperature

Option 2:
Low pO_2, high pCO_2, more H^+, higher temperature

Option 3:
High pO_2, high pCO_2, less H^+, higher temperature

Option 4:
Low pO_2, low pCO_2, more H^+, higher temperature

Correct Answer:
High pO_2, low pCO_2, less H^+, lower temperature

Solution:
Correct option is (a)

Oxyhaemoglobin is formed near alveoli. High pO_2, low pCO_2, low H^+, high pH^+. Low temperature are the conditions favourable for the same.

Q. 83 Which one of the following is an example of Hormone releasing IUD ?

Option 1:
CuT

Option 2:
LNG 20

Option 3:
Cu 7

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Option 4:
Multiload 375

Correct Answer:
LNG 20

Solution:
Correct option is (b)

LNG 20 is hormone containing IUD

Q. 84 Sphincter of oddi is present at :

Option 1:
Ileo-Caecal junction

Option 2:
Junction of hepato-pancreatic duct and duodenum

Option 3:
Gastro-oesophageal junction

Option 4:
Junction of jejunum and duodenum

Correct Answer:
Junction of hepato-pancreatic duct and duodenum

Solution:
Correct option is (b)

Junction of hepato-pancreatic duct and duodenum.

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Q. 85 Read the following statements:

(a) Metagenesis is observed in Helminths

(b) Echinoderms are triploblastic and coelomate animals

(c) Round worms have organ-system level of body organization.

(d) Comb plates present in ctenophores help in digestion.

(e) water vascular system is characterristic of Echinoderms.

Choose the **correct** answer from the options given below.

Option 1:
(c),(d) and (e) are correct

Option 2:
(a),(b) and (c) are correct

Option 3:
(a),(d) and (e) re correct

Option 4:
(b), (c) and (e) are correct

Correct Answer:
(b), (c) and (e) are correct

Solution:
Correct option is (d)

(b) Echinoderms are triploblastic and coelomate animals

(c) Round worms have organ-system level of body organization.

(e) water vascular system is characterristic of Echinoderms.

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Q. 86 Which of these is not an important component of initiation of parturition in humans?

Option 1:
Increases in estrogen and progesterone ratio

Option 2:
Synthesis of prostaglandins

Option 3:
Release of Oxytocin

Option 4:
Release of Prolactin

Correct Answer:
Increases in estrogen and progesterone ratio

Solution:
Correct option is (a)

Increases in estrogen and progesterone ratio is not an important factor for parturition.

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Q. 87 During muscular contraction which of following event occur?

(a) 'H' zone disappears

(b) 'A' band widens

(c) 'I' band reduces in width

(d) Myosine hydrolyzes ATP, releasing the ADP___and Pi

(e) Z-lines attached to actins are pulled in

Choose the correct answer from the option given below.

Option 1:
(a),(c),(d),(e) only

Option 2:
(a),(b),(c),(d) only

Option 3:
(b),(c),(d),(e) only

Option 4:
(b),(d),(e),(a) only

Correct Answer:
(a),(c),(d),(e) only

Solution:
Correct option is (a)

(a) 'H' zone disappears

(c) 'I' band reduces in width

(d) Myosine hydrolyzes ATP, releasing the ADPand Pi

(e) Z-lines attached to actins are pulled in

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Q. 88 Following are the statements with reference to 'lipids'.

(a) Lipids having only single bonds are called unsaturated fatty acids.

(b) Lecithin is a phospholipid.

(c) Trihydroxy propane is glycerol.

(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.

(e) Arachidonic acid has 16 carbon atoms.

Choose the correct answer from the options given below.

Option 1:
(a) and (b) only

Option 2:
(c) and (d) only

Option 3:
(b) and (c) only

Option 4:
(b) and (e) only

Correct Answer:
(b) and (c) only

Solution:
- **Statement (b) is correct. L**ecithin is indeed a phospholipid.
- **Statement (c) is also correct.** Trihydroxy propane is another name for glycerol, which is a common component of lipids.
- Statements (a), (d), and (e) are incorrect:
- Statement (a) is incorrect. Lipids having only single bonds are called saturated fatty acids, not unsaturated fatty acids.
- Statement (d) is incorrect. Palmitic acid has 16 carbon atoms, not 20.
- Statement (e) is incorrect. Arachidonic acid has 20 carbon atoms, not 16.
- Therefore, the correct answer is option 3.

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Q. 89 Which of the following is not a step in multiple Ovulation Embryo Transfer Technology (MOET) ?

Option 1:
Cow is administered hormone having LH like activity for super ovulation

Option 2:
Cow yeilds about 6-8 eggs at a time

Option 3:
Cow is fertilized by artificial insemination

Option 4:
Fertilized eggs are transferred to surrogate mothers at 8-32 cell stage

Correct Answer:
Cow is administered hormone having LH like activity for super ovulation

Solution:
Correct option is (a)

In this method, cow is administered with FSH like hormone, **to induce follicular maturation** and then the super ovulation is brought about. In each cycle, 6−8 eggs mature simultaneously.

Q. 90 Identify the types of cell junctions that helps to stop the leakage of the substances across a tissue and facilitation of communication with neighbouring cells via rapid transfer of ions and molecules.

Option 1:
Gap junctions and Adhering junctions, respectively.

Option 2:
Tight junctions and Gap junctions, respectively.

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Option 3:
Adhering junctions and Tight junctions, respectively.

Option 4:
Adhering junctions and Gap junctions, respectively.

Correct Answer:
Tight junctions and Gap junctions, respectively.

Solution:
Correct option is (b)

Tight Junctions $\longrightarrow$ Brings about leakage of substances.

Gap Junctions $\longrightarrow$ Communication between cytoplasm.

Q. 91 The Adenosine deaminase deficiency results into:

Option 1:
Dysfunction of Immune system

Option 2:
Parkinson's disease

Option 3:
Digestive disorder

Option 4:
Addison's disease

Correct Answer:
Dysfunction of Immune system

Solution:
Correct option is (a)

Dysfunction of Immune system

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Q. 92 **Statement I:**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

'AAA' and 'AAG both codons code for the amino acid lysine.

In the light of the above statements, choose the **correct** answer from the options given below.

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is correct but Statement II is false

Option 4:
Statement I is incorrect but Statement II is true.

Correct Answer:
Statement I is incorrect but Statement II is true.

Solution:
AUG codes for only methionine but not tryptophan. (A given codon codes for only one amino acid but not more than one amino acids.). AAG and AAA are the two codons for the amino acid Lysine.

Hence,Correct option is (d) - Statement I is incorrect but Statement II is true.

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Q. 93 Match **List - I** with **List - II**.

List - I		**List - II**	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Fibrous joints
(d)	Vertebral Column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

Option 1:
(a) (b) (c) (d)
(i) (iii) (ii) (iv)

Option 2:
(a) (b) (c) (d)
(ii) (iii) (iv) (i)

Option 3:
(a) (b) (c) (d)
(iv) (ii) (iii) (i)

Option 4:
(a) (b) (c) (d)
(iv) (iii) (ii) (i)

Correct Answer:
(a) (b) (c) (d)
(iv) (iii) (ii) (i)

Solution:
Correct option is (d)

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Scapula - Triangular flat bone

Cranium - Fibrous joints

Sternum - Flat bone

Vertebral column - Cartilaginous joints

Q. 94 Match **List - I** with **List - II**.

List - I		**List - II**	
(a)	Filariasis	(i)	Haemophilus Influenzae
(b)	Amoebiasis	(ii)	Trichophyton
(c)	Pneumonia	(iii)	Wuchereria bancrofti
(d)	Ringworm	(iv)	Entamoeba Histolytica

Choose the correct anwer from the options given below.

Option 1:
(a) (b) (c) (d)
(iv) (i) (iii) (ii)

Option 2:
(a) (b) (c) (d)
(iii) (iv) (i) (ii)

Option 3:
(a) (b) (c) (d)
(i) (ii) (iv) (iii)

Option 4:
(a) (b) (c) (d)
(ii) (iii) (i) (iv)

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Correct Answer:

(a) (b) (c) (d)

(iii) (iv) (i) (ii)

Solution:

Correct option is (b)

(a)	Filariasis	(i)	Wuchereria bancrofti
(b)	Amoebiasis	(ii)	Entamoeba Histolytica
(c)	Pneumonia	(iii)	Haemophilus Influenzae
(d)	Ringworm	(iv)	Trichophyton

Q. 95 Following are the statements about prostomium of earthworm.

(a) It serves as a covering for mouth

(b) It helps to open cracks in the soil into which it can crawl.

(c) It is one of the sensory structures.

(d) It is the first body segment.

Choose the correct answer from the options given below.

Option 1:

(a), (b) and (c) are correct

Option 2:

(a), (b) and (d) are correct

Option 3:

(a), (b), (c) and (d) are correct

Option 4:

(b) and (c) are correct

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Correct Answer:
(a), (b) and (c) are correct

Solution:
Correct option is (a)

(a) It serves as a covering for mouth

(b) It helps to open cracks in the soil into which it can crawl.

(c) It is one of the sensory structures.

Q. 96 Which of the following secretes the hormone, relaxin, during the later phase of pregnancy ?

Option 1:
Graafian follicle

Option 2:
Corpus luteum

Option 3:
Foetus

Option 4:
Uterus

Correct Answer:
Corpus luteum

Solution:
Correct option is (b)

Corpus luteum

- The **ovulation** (ovulatory phase) is followed by the **luteal phase** during which the remaining parts of the Graafian follicle transform as the corpus luteum.
- The corpus luteum secretes large amounts of progesterone which is essential for the maintenance of the endometrium.
- Such an endometrium is necessary for implantation of the fertilised ovum and other events of pregnancy.

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- During pregnancy, all events of the menstrual cycle stop and there is no menstruation.
- In the absence of fertilisation, the corpus luteum degenerates.

Q. 97 Which one of the following statements about Histones is **wrong** ?

Option 1:
Histones are organized to form a unit of 8 molecules.

Option 2:
The pH of histones is slightly acidic

Option 3:
Histones are rich in amino acids- Lysine and Arginine

Option 4:
Histones carry positive charge in the side chain.

Correct Answer:
The pH of histones is slightly acidic

Solution:
Correct option is (b)

Histones are basic proteins

Q. 98 **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the **correct** answer from the options given below.

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Option 1:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Option 2:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

Option 3:
(A) is true but **(R)** is false

Option 4:
(A) is false but **(R)** is true

Correct Answer:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Solution:
The correct option is (a)

Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

Q. 99 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Allen's Rule	(i)	Kangaroo rat
(b)	Physiological adaption	(ii)	Desert lizard
(c)	Behavioural adaption	(iii)	Marine fish at depth
(d)	Biochemical adaption	(iv)	Polar seal

Choose the **correct** answer from the options given below.

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Option 1:

(a) (b) (c) (d)

(iv) (ii) (iii) (i)

Option 2:

(a) (b) (c) (d)

(iv) (i) (iii) (ii)

Option 3:

(a) (b) (c) (d)

(iv) (i) (ii) (iii)

Option 4:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Correct Answer:

(a) (b) (c) (d)

(iv) (i) (ii) (iii)

Solution:

Correct option is (c)

(a)	Allen's Rule	(i)	Polar seal
(b)	Physiological adaption	(ii)	Kangaroo rat
(c)	Behavioural adaption	(iii)	Desert Lizard
(d)	Biochemical adaption	(iv)	Marine fish at depth

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Q. 100 Match **List-I** with **List - II**.

List - I		**List - II**	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excrssive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

Option 1:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Option 2:

(a) (b) (c) (d)

(iii) (ii) (i) (iv)

Option 3:

(a) (b) (c) (d)

(ii) (i) (iv) (iii)

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Option 4:

(a) (b) (c) (d)

(i) (iv) (iii) (ii)

Correct Answer:

(a) (b) (c) (d)

(iv) (iii) (ii) (i)

Solution:

Correct option is (a)

(a)	Adaptive radiation	(i)	Darwin Finches
(b)	Convergent evolution	(ii)	Wings of Butterfly and Bird
(c)	Divergent evolution	(iii)	Bones of forelimbs in Man and Whale
(d)	Evolution by anthropogenic action	(iv)	Selection of resistant varieties due to excrssive use of herbicides and pesticides

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